SAP HANA on NetApp All Flash FAS Systems with NFS Configuration Guide

NetApp Solutions

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TR-4435: SAP HANA on NetApp AFF Systems with NFS - Configuration Guide

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The NetApp AFF system product family has been certified for use with SAP HANA in tailored data center integration (TDI) projects. The certified enterprise storage system is characterized by the NetApp ONTAP software.

This certification is valid for the following models:

- AFF A220, AFF A250, AFF A300, AFF A320, AFF A400, AFF A700s, AFF A700, AFF A800

A complete list of NetApp certified storage solutions for SAP HANA can be found at the Certified and supported SAP HANA hardware directory.

This document describes the ONTAP configuration requirements for the NFS protocol version 3 (NFSv3) or the NFS protocol version 4 (NFSv4.0 and NFSv4.1).

For the remainder of this document, NFSv4 refers to both NFSv4.0 and NFSv4.1.

> The configuration described in this paper is necessary to achieve the required SAP HANA KPIs and the best performance for SAP HANA. Changing any settings or using features not listed herein might cause performance degradation or unexpected behavior and should only be done if advised by NetApp support.

The configuration guides for NetApp AFF systems using FCP and for FAS systems using NFS or FCP can be found at the following links:

- SAP HANA on NetApp FAS Systems with Fibre Channel Protocol
- SAP HANA on NetApp FAS Systems with NFS
- SAP HANA on NetApp AFF Systems with Fibre Channel Protocol

The following table shows the supported combinations for NFS versions, NFS locking, and the required isolation implementations, depending on the SAP HANA database configuration.

For SAP HANA single-host systems or multiple hosts that do not use Host Auto-Failover, NFSv3 and NFSv4 are supported.

For SAP HANA multiple host systems with Host Auto-Failover, NetApp only supports NFSv4, while using NFSv4 locking as an alternative to a server-specific STONITH (SAP HANA HA/DR provider) implementation.

SAP HANA	NFS version	NFS locking	SAP HANA HA/DR provider
SAP HANA single host, multiple hosts without Host Auto-Failover	NFSv3	Off	n/a
	NFSv4	On	n/a
SAP HANA multiple hosts using Host Auto-Failover	NFSv3	Off	Server-specific STONITH implementation mandatory

SAP HANA	NFS version	NFS locking	SAP HANA HA/DR provider
	NFSv4	On	Not required

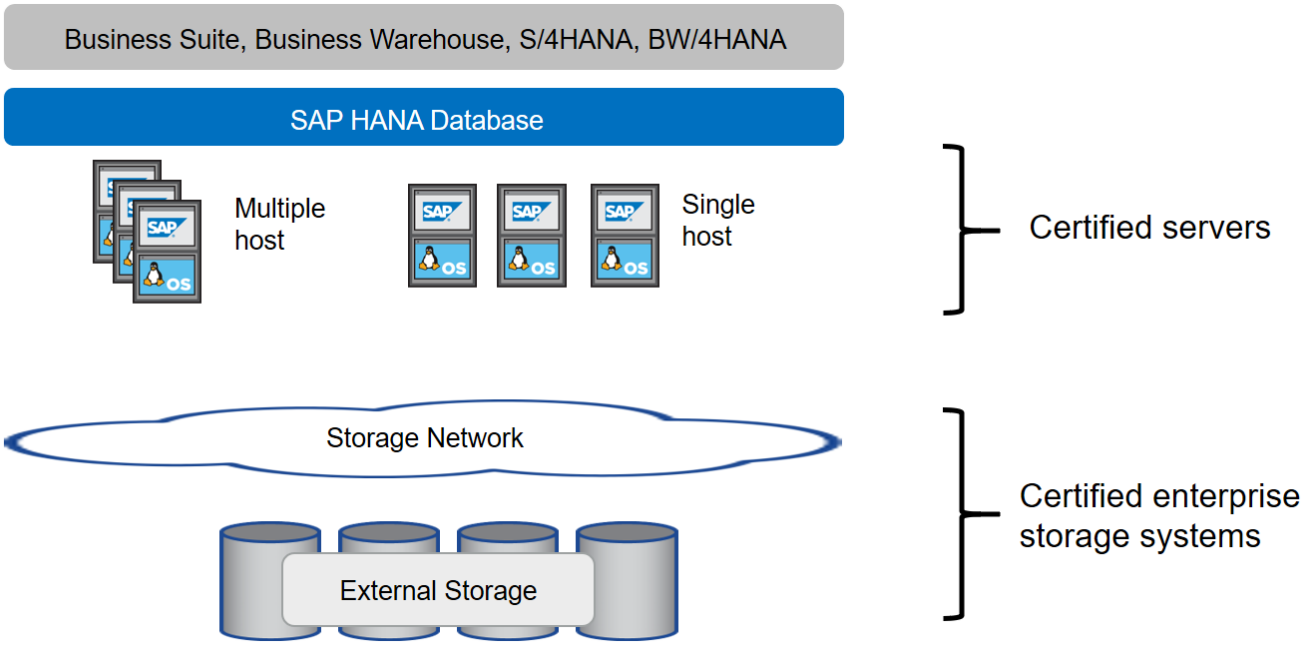
A server-specific STONITH implementation is not part of this guide. Contact your server vendor for such an implementation.

This document covers configuration recommendations for SAP HANA running on physical servers and on virtual servers that use VMware vSphere.

See the relevant SAP notes for operating system configuration guidelines and HANA-specific Linux kernel dependencies. For more information, see SAP note 2235581: SAP HANA Supported Operating Systems.

SAP HANA tailored data center integration

NetApp AFF storage controllers are certified in the SAP HANA TDI program using both NFS (NAS) and FC (SAN) protocols. They can be deployed in any of the current SAP HANA scenarios, such as SAP Business Suite on HANA, S/4HANA, BW/4HANA, or SAP Business Warehouse on HANA in either single-host or multiple-host configurations. Any server that is certified for use with SAP HANA can be combined with NetApp certified storage solutions. See the following figure for an architecture overview of SAP HANA TDI.



For more information regarding the prerequisites and recommendations for producti SAP HANA systems, see the following resources:

- SAP HANA Tailored Data Center Integration Frequently Asked Questions
- SAP HANA Storage Requirements

SAP HANA using VMware vSphere

There are several options for connecting storage to virtual machines (VMs). The preferred option is to connect the storage volumes with NFS directly out of the guest operating system. Using this option, the configuration of hosts and storage does not differ between physical hosts and VMs.

NFS datastores and VVOL datastores with NFS are supported as well. For both options, only one SAP HANA data or log volume must be stored within the datastore for production use cases. In addition, Snapshot-based backup and recovery orchestrated by NetApp SnapCenter and solutions based on this, such as SAP System cloning, cannot be implemented.

This document describes the recommended setup with direct NFS mounts from the guest OS.

For more information about using vSphere with SAP HANA, see the following links:

- SAP HANA on VMware vSphere - Virtualization - Community Wiki
- Best Practices and Recommendations for Scale-Up Deployments of SAP HANA on VMware vSphere
- Best Practices and Recommendations for Scale-Out Deployments of SAP HANA on VMware vSphere
- 2161991 - VMware vSphere configuration guidelines - SAP ONE Support Launchpad (Login required)

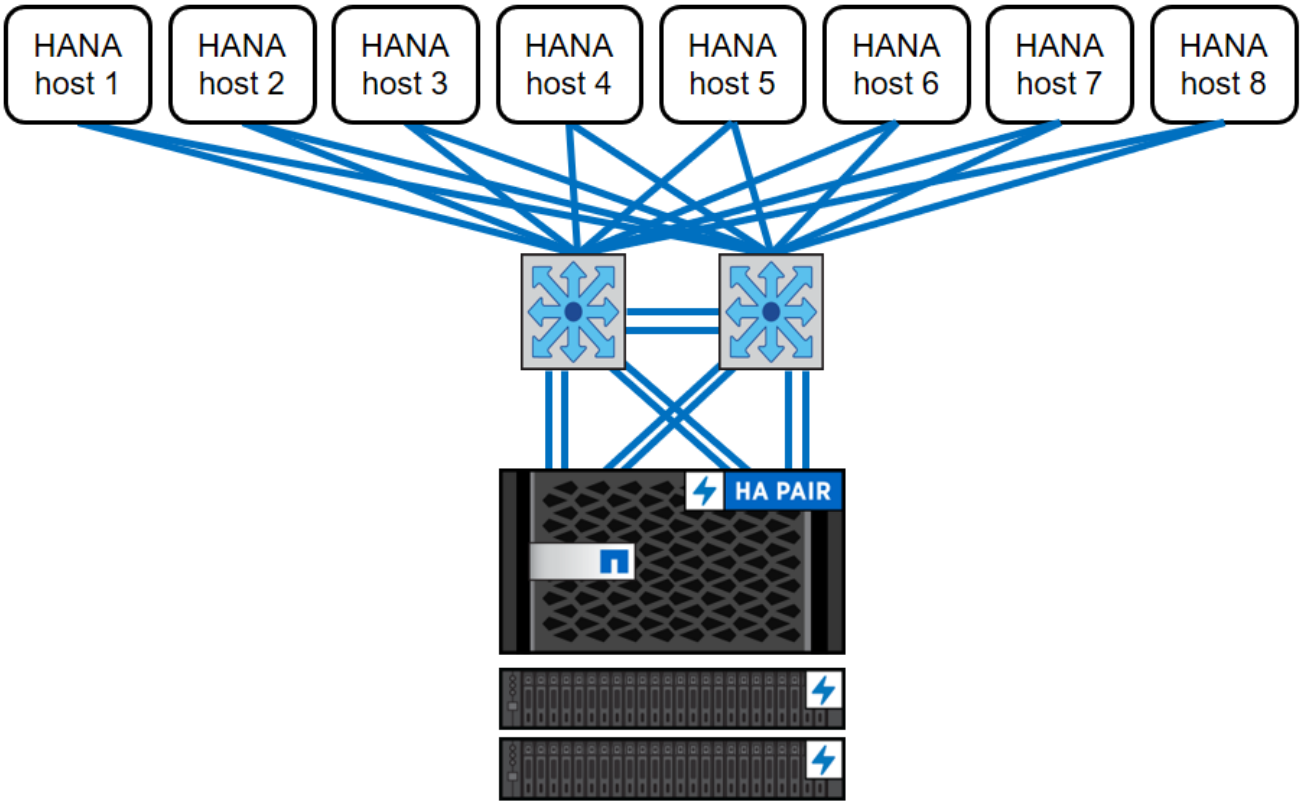
Architecture

SAP HANA hosts are connected to storage controllers by using a redundant 10GbE or faster network infrastructure. Data communication between SAP HANA hosts and storage controllers is based on the NFS protocol. A redundant switching infrastructure is required to provide fault-tolerant SAP HANA host-to-storage connectivity in case of switch or network interface card (NIC) failure.

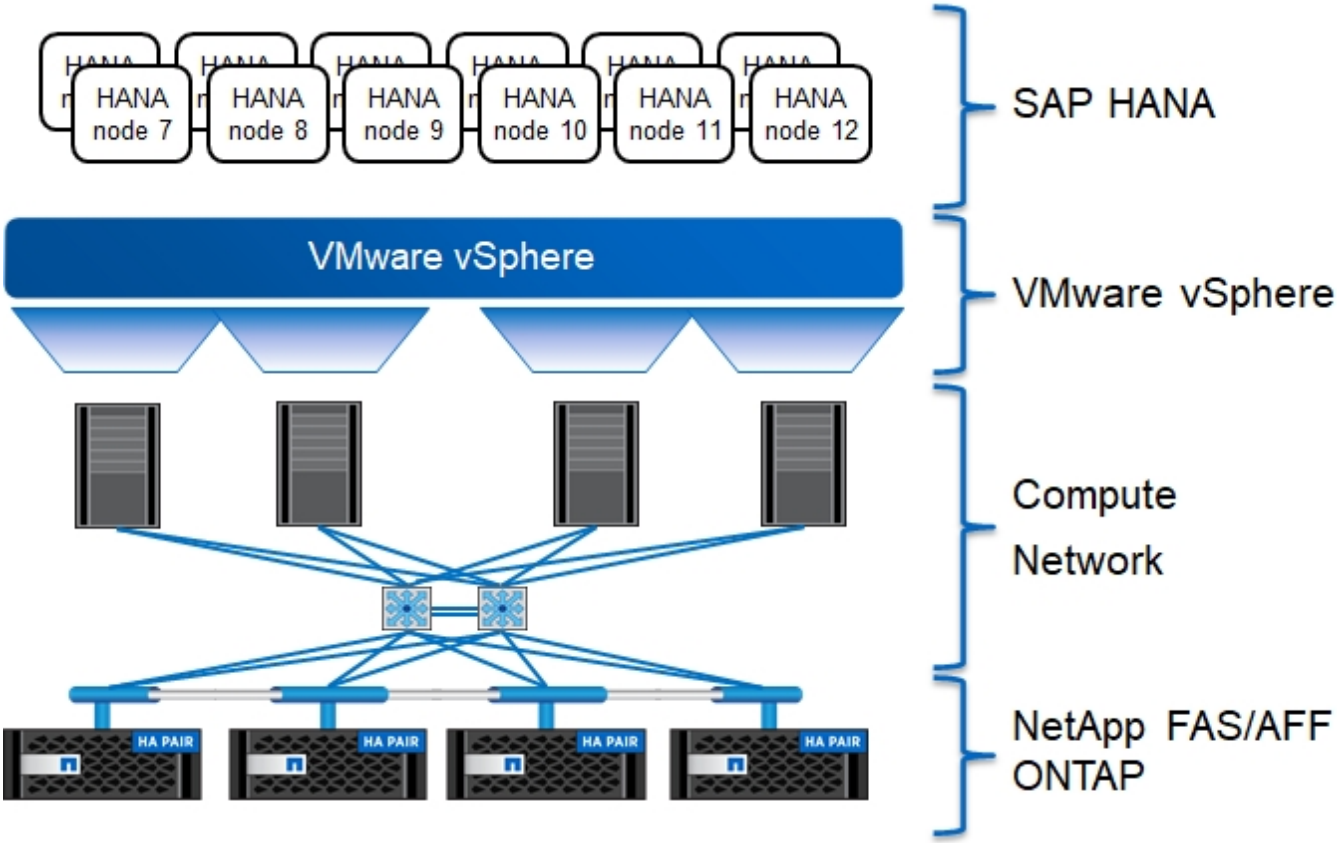
The switches might aggregate individual port performance with port channels in order to appear as a single logical entity at the host level.

Different models of the AFF system product family can be mixed and matched at the storage layer to allow for growth and differing performance and capacity needs. The maximum number of SAP HANA hosts that can be attached to the storage system is defined by the SAP HANA performance requirements and the model of NetApp controller used. The number of required disk shelves is only determined by the capacity and performance requirements of the SAP HANA systems.

The following figure shows an example configuration with eight SAP HANA hosts attached to a storage high availability (HA) pair.



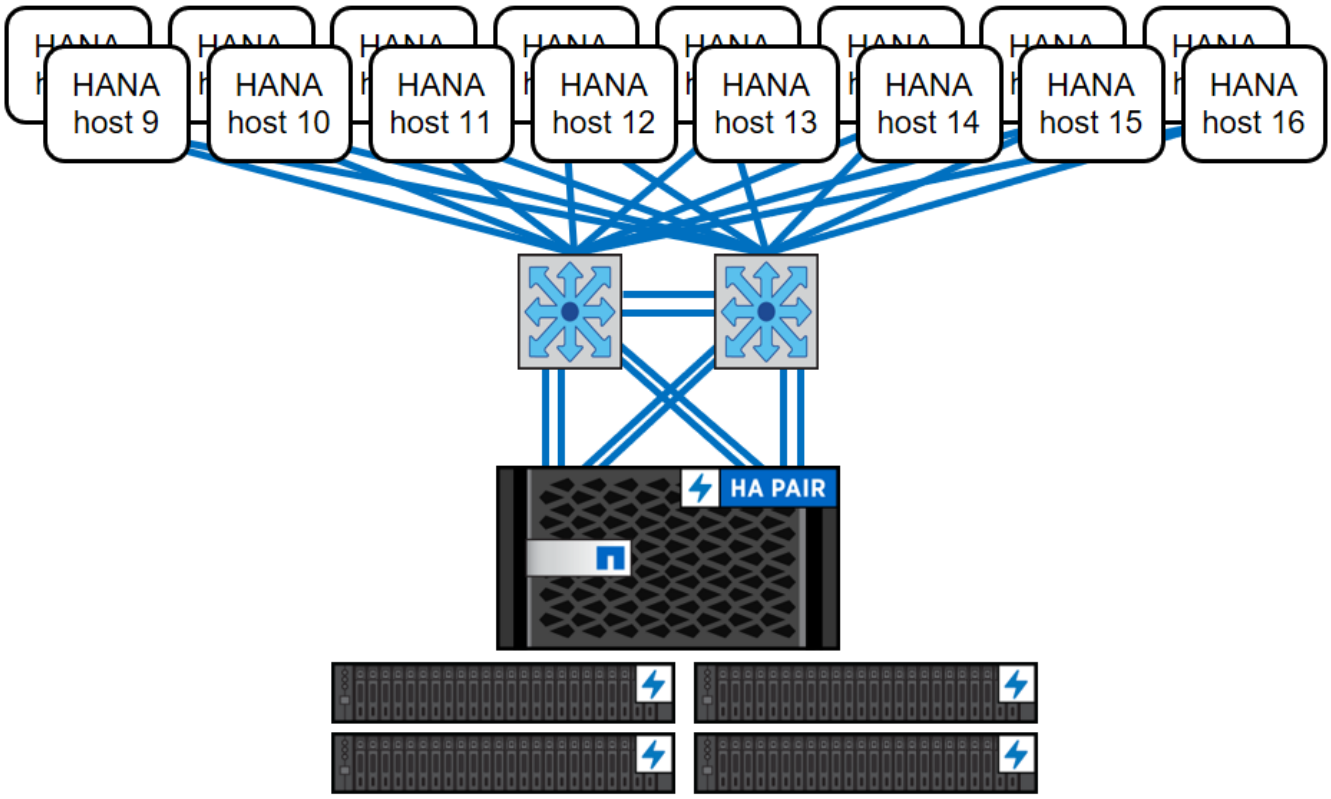
The following figure shows an example of using VMware vSphere as a virtualization layer.



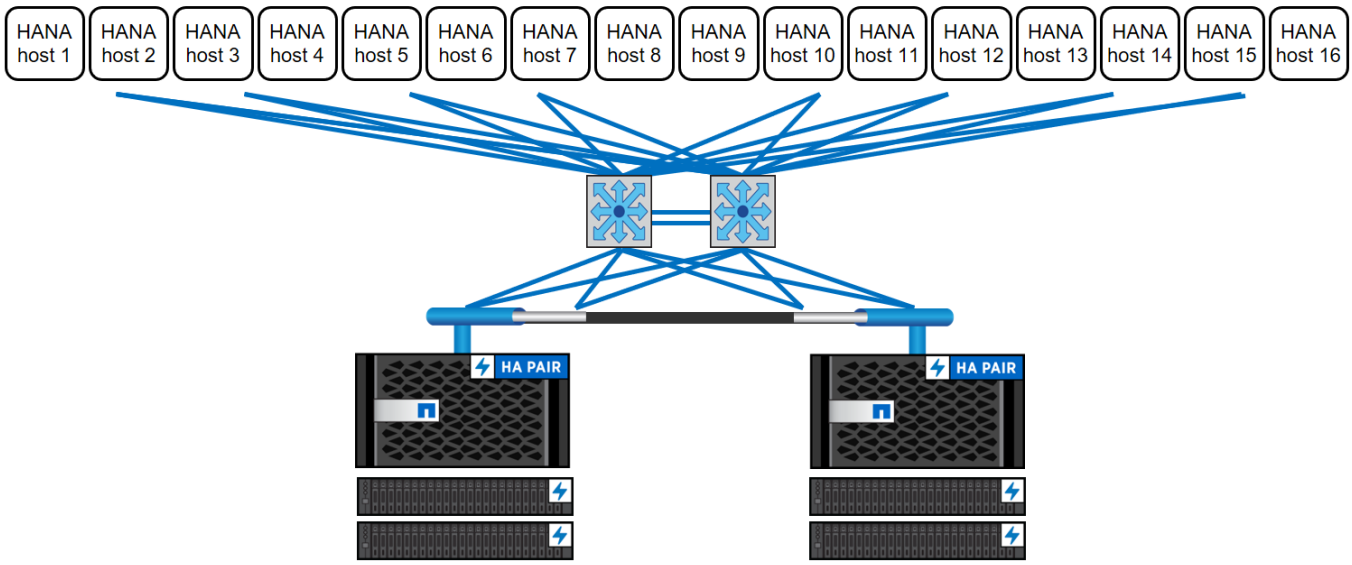
The architecture can be scaled in two dimensions:

- By attaching additional SAP HANA hosts and storage capacity to the existing storage, if the storage controllers provide enough performance to meet the current SAP HANA key performance indicators (KPIs).
- By adding more storage systems with additional storage capacity for the additional SAP HANA hosts

The following figure shows an example configuration in which more SAP HANA hosts are attached to the storage controllers. In this example, more disk shelves are necessary to fulfill the capacity and performance requirements of the 16 SAP HANA hosts. Depending on the total throughput requirements, you must add additional 10GbE or faster connections to the storage controllers.



Independent of the deployed AFF system, the SAP HANA landscape can also be scaled by adding any of the certified storage controllers to meet the desired node density, as shown in the following figure.



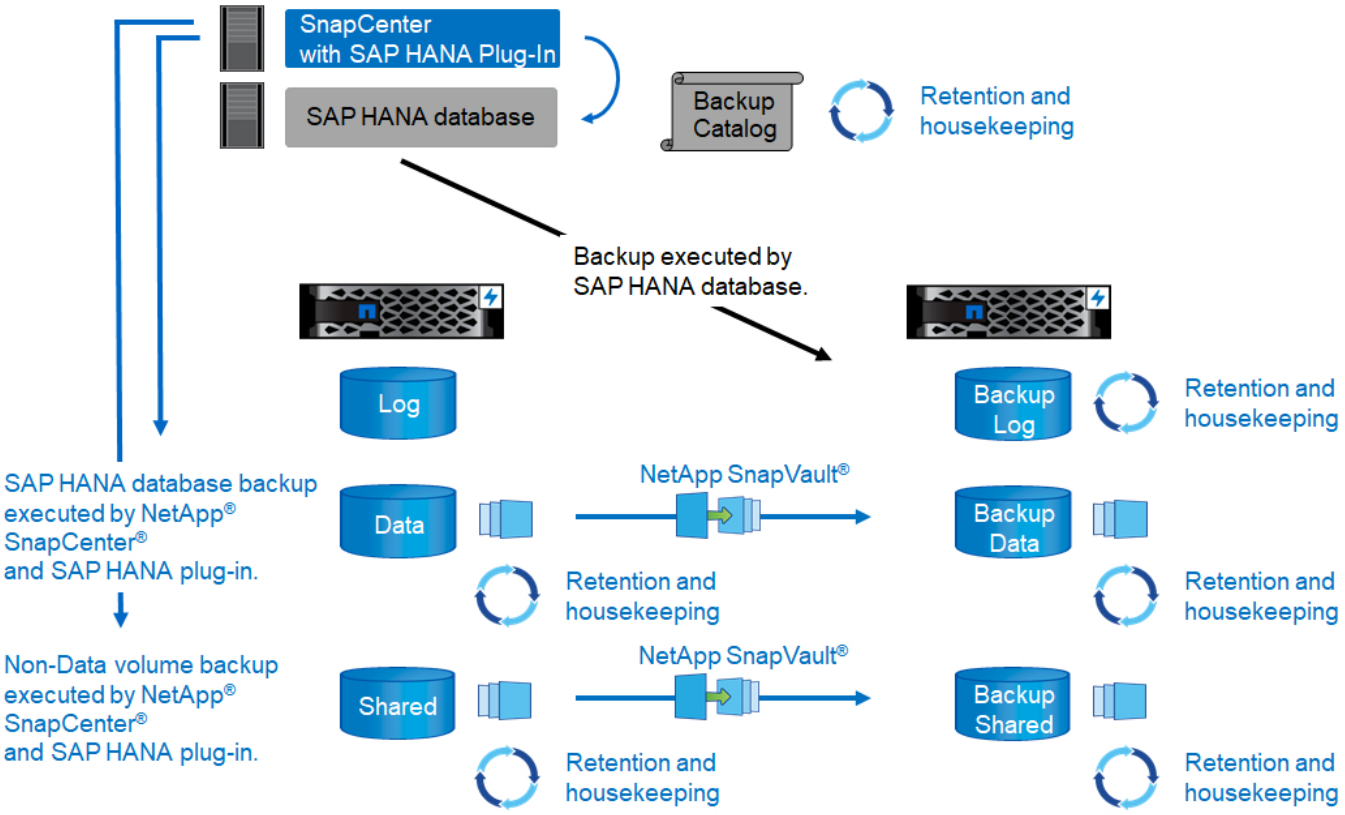
SAP HANA backup

The ONTAP software present on all NetApp storage controllers provides a built-in mechanism to back up SAP HANA databases while in operation with no effect on performance. Storage-based NetApp Snapshot backups are a fully supported and integrated backup solution available for SAP HANA single containers and for SAP HANA Multitenant Database Containers (MDC) systems with a single tenant or multiple tenants.

Storage-based Snapshot backups are implemented by using the NetApp SnapCenter plug-in for SAP HANA. This allows users to create consistent storage-based Snapshot backups by using the interfaces provided natively by SAP HANA databases. SnapCenter registers each of the Snapshot backups into the SAP HANA backup catalog. Therefore, the backups taken by SnapCenter are visible within SAP HANA Studio and Cockpit where they can be selected directly for restore and recovery operations.

NetApp SnapMirror technology enables Snapshot copies that were created on one storage system to be replicated to a secondary backup storage system that is controlled by SnapCenter. Different backup retention policies can then be defined for each of the backup sets on the primary storage and for the backup sets on the secondary storage systems. The SnapCenter Plug-in for SAP HANA automatically manages the retention of Snapshot copy-based data backups and log backups, including the housekeeping of the backup catalog. The SnapCenter Plug-in for SAP HANA also allows the execution of a block integrity check of the SAP HANA database by executing a file-based backup.

The database logs can be backed up directly to the secondary storage by using an NFS mount, as shown in the following figure.



Storage-based Snapshot backups provide significant advantages compared to conventional file-based backups. These advantages include, but are not limited to, the following:

- Faster backup (a few minutes)
- Reduced recovery time objective (RTO) due to a much faster restore time on the storage layer (a few minutes) as well as more frequent backups
- No performance degradation of the SAP HANA database host, network, or storage during backup and recovery operations
- Space-efficient and bandwidth-efficient replication to secondary storage based on block changes

For detailed information about the SAP HANA backup and recovery solution see TR-4614: SAP HANA Backup and Recovery with SnapCenter.

SAP HANA disaster recovery

SAP HANA disaster recovery (DR) can be done either on the database layer by using SAP HANA system replication or on the storage layer by using storage replication technologies. The following section provides an overview of disaster recovery solutions based on storage replication.

For detailed information about SAP HANA disaster recovery solutions, see TR-4646: SAP HANA Disaster Recovery with Storage Replication.

Storage replication based on SnapMirror

The following figure shows a three-site disaster recovery solution using synchronous SnapMirror replication to the local DR datacenter and asynchronous SnapMirror to replicate the data to the remote DR datacenter.

Data replication using synchronous SnapMirror provides an RPO of zero. The distance between the primary

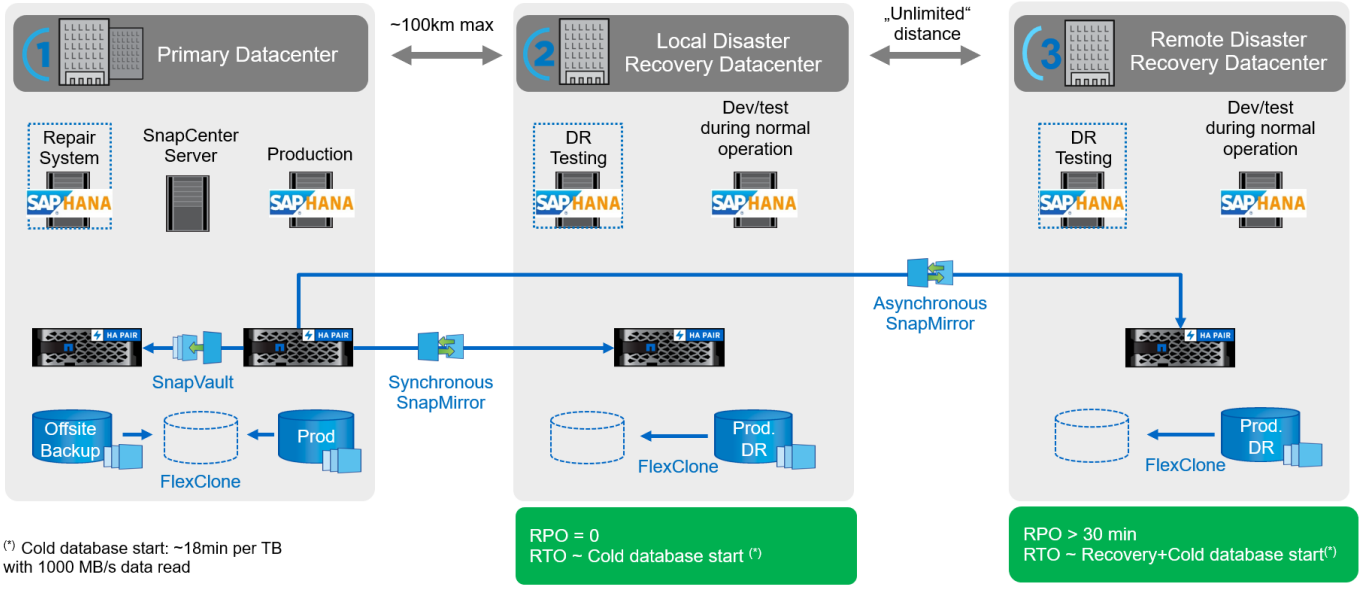
and the local DR datacenter is limited to around 100km.

Protection against failures of both the primary and the local DR site is performed by replicating the data to a third remote DR datacenter using asynchronous SnapMirror. The RPO depends on the frequency of replication updates and how fast they can be transferred. In theory, the distance is unlimited, but the limit depends on the amount of data that must be transferred and the connection that is available between the data centers. Typical RPO values are in the range of 30 minutes to multiple hours.

The RTO for both replication methods primarily depends on the time needed to start the HANA database at the DR site and load the data into memory. With the assumption that the data is read with a throughput of 1000MBps, loading 1TB of data would take approximately 18 minutes.

The servers at the DR sites can be used as dev/test systems during normal operation. In the case of a disaster, the dev/test systems would need to be shut down and started as DR production servers.

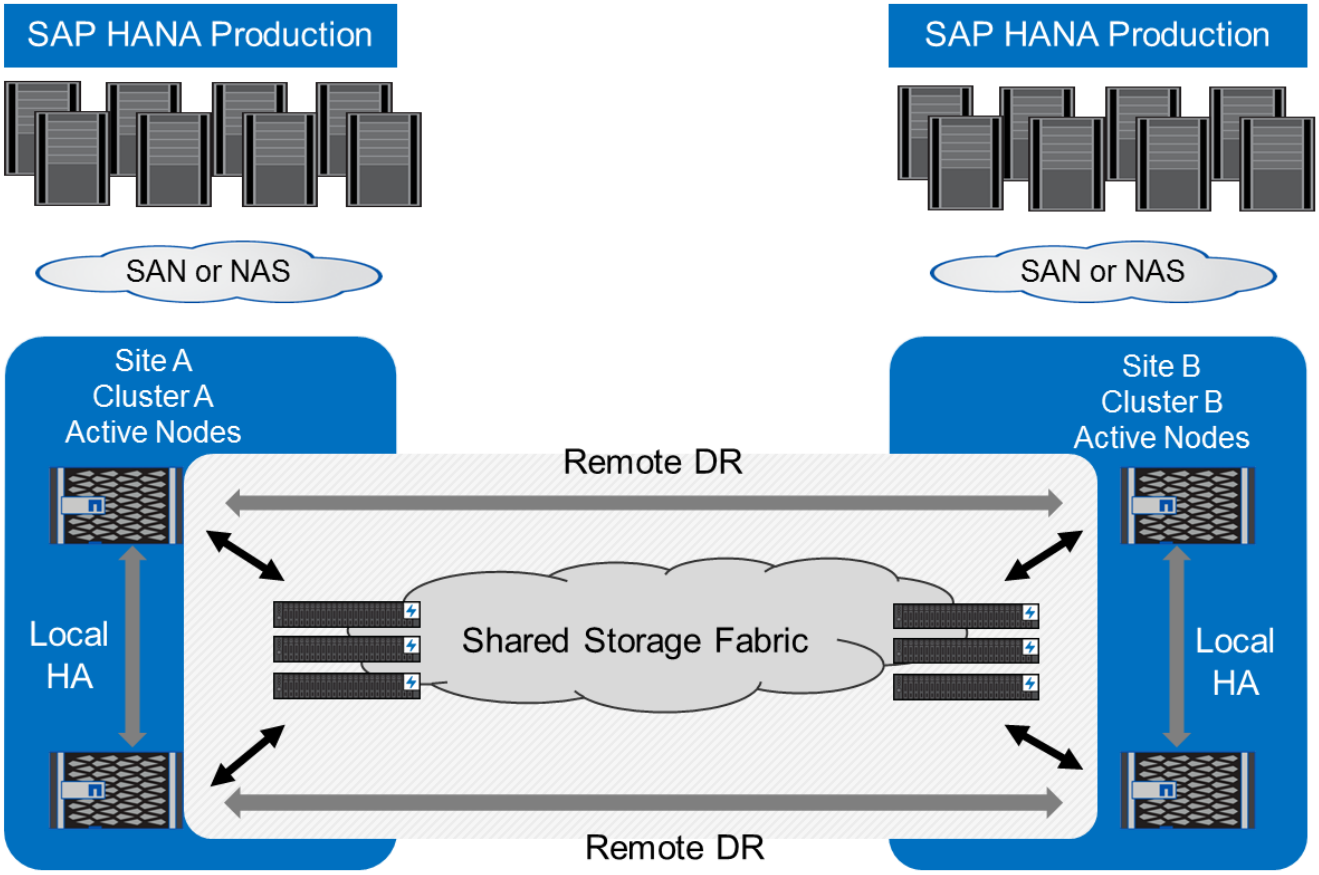
Both replication methods allow to you execute DR workflow testing without influencing the RPO and RTO. FlexClone volumes are created on the storage and are attached to the DR testing servers.



Synchronous replication offers StrictSync mode. If the write to secondary storage is not completed for any reason, the application I/O fails, thereby ensuring that the primary and secondary storage systems are identical. Application I/O to the primary resumes only after the SnapMirror relationship returns to the InSync status. If the primary storage fails, application I/O can be resumed on the secondary storage after failover with no loss of data. In StrictSync mode, the RPO is always zero.

Storage replication based on MetroCluster

The following figure shows a high-level overview of the solution. The storage cluster at each site provides local high availability and is used for the production workload. The data of each site is synchronously replicated to the other location and is available in case of disaster failover.



Storage sizing

The following section provides an overview of the required performance and capacity considerations needed for sizing a storage system for SAP HANA.

> Contact NetApp or your NetApp partner sales representative to assist you in creating a properly sized storage environment.

Performance considerations

SAP has defined a static set of storage KPIs. These KPIs are valid for all production SAP HANA environments independent of the memory size of the database hosts and the applications that use the SAP HANA database. These KPIs are valid for single-host, multiple-host, Business Suite on HANA, Business Warehouse on HANA, S/4HANA, and BW/4HANA environments. Therefore, the current performance sizing approach depends on only the number of active SAP HANA hosts that are attached to the storage system.

> Storage performance KPIs are only mandated for production SAP HANA systems, but you can implement them in for all HANA system.

SAP delivers a performance test tool that must be used to validate the storage system's performance for active SAP HANA hosts attached to the storage.

NetApp tested and predefined the maximum number of SAP HANA hosts that can be attached to a specific storage model while still fulfilling the required storage KPIs from SAP for production-based SAP HANA systems.

The maximum number of SAP HANA hosts that can be run on a disk shelf and the minimum number of SSDs required per SAP HANA host were determined by running the SAP performance test tool. This test does not consider the actual storage capacity requirements of the hosts. You must also calculate the capacity requirements to determine the actual storage configuration needed.

SAS disk shelf

With the 12Gb serial-attached SCSI (SAS) disk shelf (DS224C), performance sizing is performed by using the following fixed disk-shelf configurations:

- Half-loaded disk shelves with 12 SSDs
- Fully loaded disk shelves with 24 SSDs

> Both configurations use Advanced Disk Partitioning (ADPv2). A half-loaded disk shelf supports up to nine SAP HANA hosts, whereas a fully loaded shelf supports up to 14 hosts in a single disk shelf. The SAP HANA hosts must be equally distributed between both storage controllers. The same applies to the internal disks of an AFF A700s system. The DS224C disk shelf must be connected using 12Gb SAS to support the number of SAP HANA hosts.

The 6Gb SAS disk shelf (DS2246) supports a maximum of four SAP HANA hosts. The SSDs and the SAP HANA hosts must be equally distributed between both storage controllers.

The following table summarizes the supported number of SAP HANA hosts per disk shelf.

	6Gb SAS shelves (DS2246)Fully loaded with 24 SSDs	**12Gb SAS shelves (DS224C)Half loaded with 12 SSDs and ADPv2**	**12Gb SAS shelves (DS224C)Fully loaded with 24 SSDs and ADPv2**
Maximum number of SAP HANA hosts per disk shelf	4	9	14

> This calculation is independent of the storage controller used. Adding more disk shelfs do not increase the maximum amount of SAP HANA hosts a storage controller can support.

NS224 NVMe shelf

The minimum number of 12 NVMe SSDs for the first shelf supports up to 16 SAP HANA hosts. A fully populated shelf (24 SSDs) supports up to 34 SAP HANA hosts. The same applies to the internal disks of an AFF A800 system.

> Adding more disk shelfs does not increase the maximum amount of SAP HANA hosts a storage controller can support.

Mixed workloads

SAP HANA and other application workloads running on the same storage controller or in the same storage aggregate are supported. However, it is a NetApp best practice to separate SAP HANA workloads from all other application workloads.

You might decide to deploy SAP HANA workloads and other application workloads on either the same storage

controller or the same aggregate. If so, you must make sure that adequate performance is available for SAP HANA within the mixed workload environment. NetApp also recommends that you use quality of service (QoS) parameters to regulate the effect these other applications could have on SAP HANA applications and to guarantee throughput for SAP HANA applications.

The SAP performance test tool must be used to check if additional SAP HANA hosts can be run on an existing storage controller that is already in use for other workloads. SAP application servers can be safely placed on the same storage controller and/or aggregate as the SAP HANA databases.

Capacity considerations

A detailed description of the capacity requirements for SAP HANA is in the SAP HANA Storage Requirements white paper.

> The capacity sizing of the overall SAP landscape with multiple SAP HANA systems must be determined by using SAP HANA storage sizing tools from NetApp. Contact NetApp or your NetApp partner sales representative to validate the storage sizing process for a properly sized storage environment.

Configuring the performance test tool

Starting with SAP HANA 1.0 SPS10, SAP introduced parameters to adjust the I/O behavior and optimize the database for the file and storage system used. These parameters must also be set for the performance test tool from SAP when storage performance is being tested with the SAP performance test tool.

NetApp conducted performance tests to define the optimal values. The following table lists the parameters that must be set within the configuration file of the SAP performance test tool.

Parameter	Value
max_parallel_io_requests	128
async_read_submit	on
async_write_submit_active	on
async_write_submit_blocks	all

For more information about the configuration of the different SAP test tools, see SAP note 1943937 for HWCCT (SAP HANA 1.0) and SAP note 2493172 for HCMT/HCOT (SAP HANA 2.0).

The following example shows how variables can be set for the HCMT/HCOT execution plan.

```
…{
         "Comment": "Log Volume: Controls whether read requests are
submitted asynchronously, default is 'on'",
         "Name": "LogAsyncReadSubmit",
         "Value": "on",
         "Request": "false"
      },
      {
         "Comment": "Data Volume: Controls whether read requests are
submitted asynchronously, default is 'on'",
```

```
         "Name": "DataAsyncReadSubmit",
         "Value": "on",
         "Request": "false"
      },
      {
         "Comment": "Log Volume: Controls whether write requests can be
submitted asynchronously",
         "Name": "LogAsyncWriteSubmitActive",
         "Value": "on",
         "Request": "false"
      },
      {
         "Comment": "Data Volume: Controls whether write requests can be
submitted asynchronously",
         "Name": "DataAsyncWriteSubmitActive",
         "Value": "on",
         "Request": "false"
      },
      {
         "Comment": "Log Volume: Controls which blocks are written
asynchronously. Only relevant if AsyncWriteSubmitActive is 'on' or 'auto'
and file system is flagged as requiring asynchronous write submits",
         "Name": "LogAsyncWriteSubmitBlocks",
         "Value": "all",
         "Request": "false"
      },
      {
         "Comment": "Data Volume: Controls which blocks are written
asynchronously. Only relevant if AsyncWriteSubmitActive is 'on' or 'auto'
and file system is flagged as requiring asynchronous write submits",
         "Name": "DataAsyncWriteSubmitBlocks",
         "Value": "all",
         "Request": "false"
      },
      {
         "Comment": "Log Volume: Maximum number of parallel I/O requests
per completion queue",
         "Name": "LogExtMaxParallelIoRequests",
         "Value": "128",
         "Request": "false"
      },
      {
         "Comment": "Data Volume: Maximum number of parallel I/O requests
per completion queue",
         "Name": "DataExtMaxParallelIoRequests",
         "Value": "128",
```

```
         "Request": "false"
      }, …
```

These variables must be used for the test configuration. This is usually the case with the predefined execution plans SAP delivers with the HCMT/HCOT tool. The following example for a 4k log write test is from an execution plan.

```
…
      {
         "ID": "D664D001-933D-41DE-A904F304AEB67906",
         "Note": "File System Write Test",
         "ExecutionVariants": [
            {
               "ScaleOut": {
                  "Port": "${RemotePort}",
                  "Hosts": "${Hosts}",
                  "ConcurrentExecution": "${FSConcurrentExecution}"
               },
               "RepeatCount": "${TestRepeatCount}",
               "Description": "4K Block, Log Volume 5GB, Overwrite",
               "Hint": "Log",
               "InputVector": {
                  "BlockSize": 4096,
                  "DirectoryName": "${LogVolume}",
                  "FileOverwrite": true,
                  "FileSize": 5368709120,
                  "RandomAccess": false,
                  "RandomData": true,
                  "AsyncReadSubmit": "${LogAsyncReadSubmit}",
                  "AsyncWriteSubmitActive":
"${LogAsyncWriteSubmitActive}",
                  "AsyncWriteSubmitBlocks":
"${LogAsyncWriteSubmitBlocks}",
                  "ExtMaxParallelIoRequests":
"${LogExtMaxParallelIoRequests}",
                  "ExtMaxSubmitBatchSize": "${LogExtMaxSubmitBatchSize}",
                  "ExtMinSubmitBatchSize": "${LogExtMinSubmitBatchSize}",
                  "ExtNumCompletionQueues":
"${LogExtNumCompletionQueues}",
                  "ExtNumSubmitQueues": "${LogExtNumSubmitQueues}",
                  "ExtSizeKernelIoQueue": "${ExtSizeKernelIoQueue}"
               }
            }, …
```

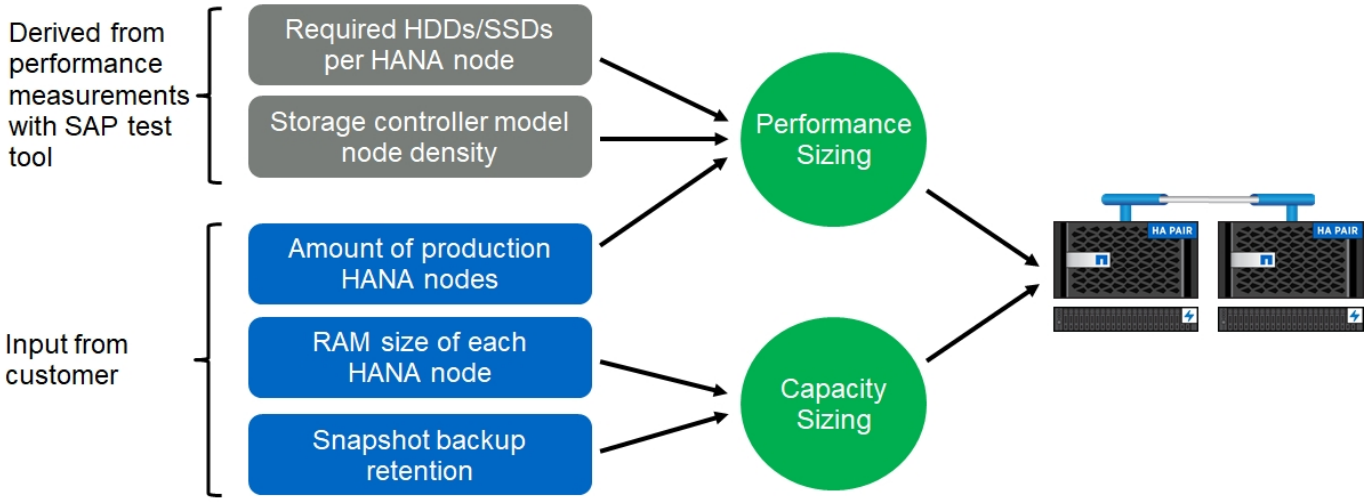
Storage sizing process overview

The number of disks per HANA host and the SAP HANA host density for each storage model were determined with performance test tool.

The sizing process requires details such as the number of production and nonproduction SAP HANA hosts, the RAM size of each host, and backup retention of the storage-based Snapshot copies. The number of SAP HANA hosts determines the storage controller and the number of disks required.

The size of the RAM, net data size on the disk of each SAP HANA host, and the Snapshot copy backup retention period are used as inputs during capacity sizing.

The following figure summarizes the sizing process.



Overview

The following sections provide SAP HANA infrastructure setup and configuration guidelines.

Network setup

Use the following guidelines when configuring the network:

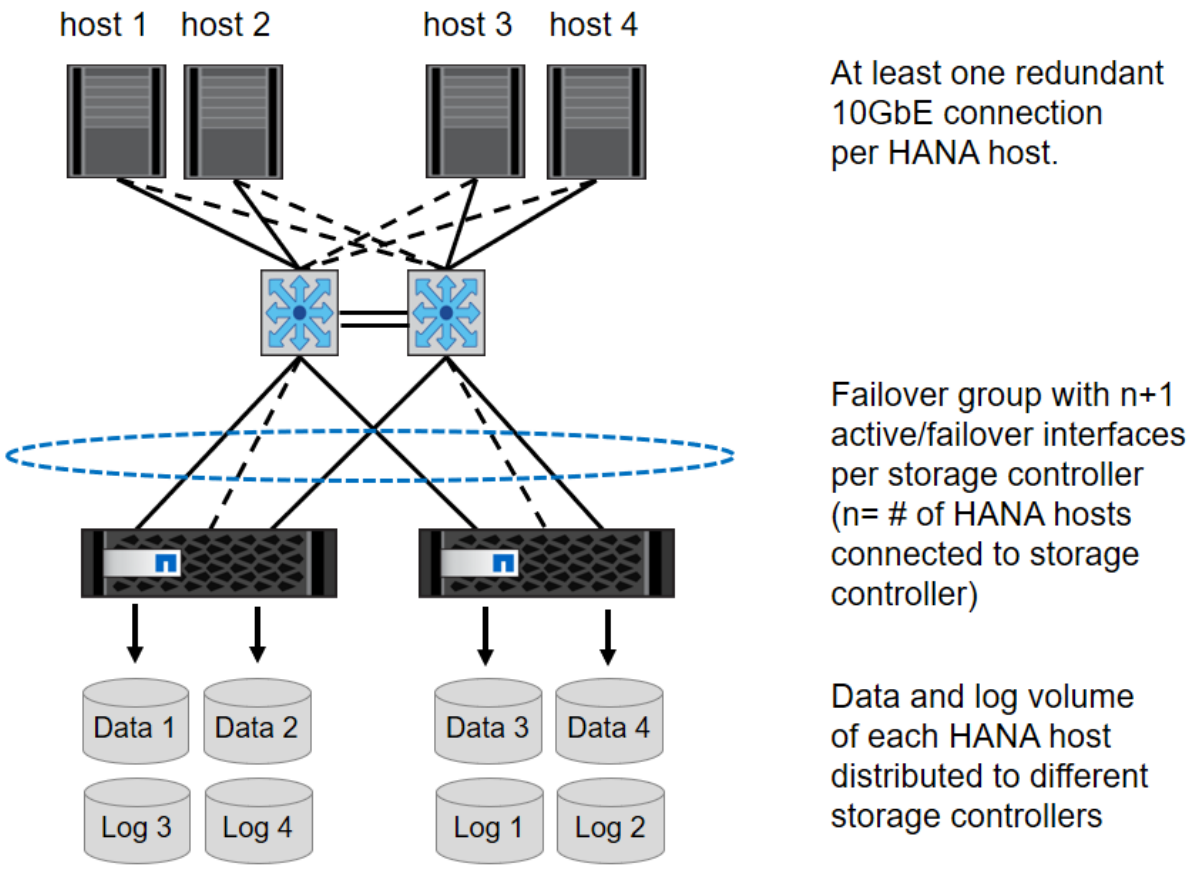
- A dedicated storage network must be used to connect the SAP HANA hosts to the storage controllers with a 10GbE or faster network.
- Use the same connection speed for storage controllers and SAP HANA hosts. If this is not possible, ensure that the network components between the storage controllers and the SAP HANA hosts are able to handle different speeds. For example, you must provide enough buffer space to allow speed negotiation at the NFS level between storage and hosts. Network components are usually switches, but other components within blade chassis, such as the back plane, must be considered as well.
- Disable flow control on all physical ports used for storage traffic on the storage network switch and host layer.
- Each SAP HANA host must have a redundant network connection with a minimum of 10Gb of bandwidth.
- Jumbo frames with a maximum transmission unit (MTU) size of 9,000 must be enabled on all network components between the SAP HANA hosts and the storage controllers.
- In a VMware setup, dedicated VMXNET3 network adapters must be assigned to each running virtual machine. Check the relevant papers mentioned in “Introduction” for further requirements.

- To avoid interference between each other, use separate network/IO paths for the log and data area.

The following figure shows an example with four SAP HANA hosts attached to a storage controller HA pair using a 10GbE network. Each SAP HANA host has an active-passive connection to the redundant fabric.

At the storage layer, four active connections are configured to provide 10Gb throughput for each SAP HANA host. In addition, one spare interface is configured on each storage controller.

At the storage layer, a broadcast domain with an MTU size of 9000 is configured, and all required physical interfaces are added to this broadcast domain. This approach automatically assigns these physical interfaces to the same failover group. All logical interfaces (LIFs) that are assigned to these physical interfaces are added to this failover group.



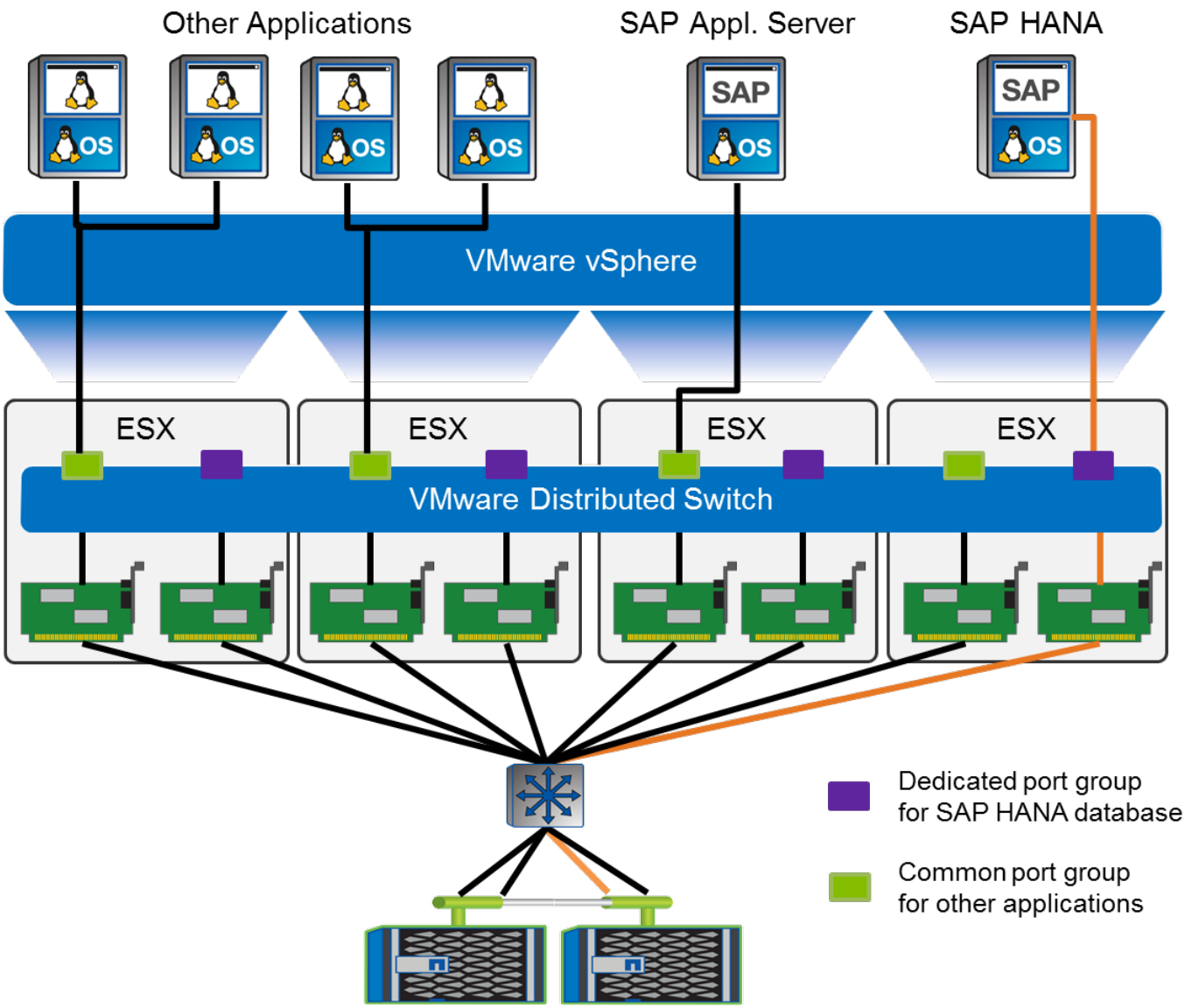
In general, it is also possible to use HA interface groups on the servers (bonds) and the storage systems (for example, Link Aggregation Control Protocol [LACP] and ifgroups). With HA interface groups, verify that the load is equally distributed between all interfaces within the group. The load distribution depends on the functionality of the network switch infrastructure.

> Depending on the number of SAP HANA hosts and the connection speed used, different numbers of active physical ports are needed. For details, see the section "LIF configuration".

VMware-specific network setup

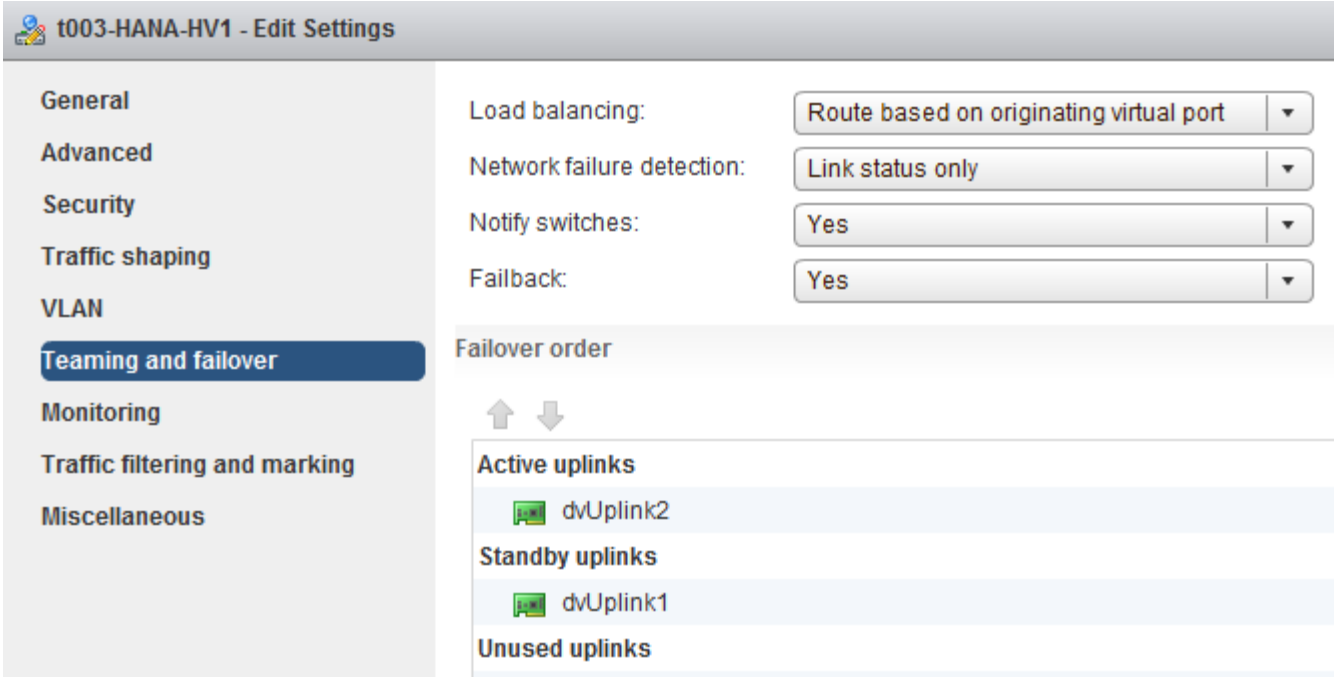
Proper network design and configuration are crucial because all data for SAP HANA instances, including performance-critical data and log volumes for the database, is provided through NFS in this solution. A dedicated storage network is used to separate the NFS traffic from communication and user access traffic

between SAP HANA nodes. Each SAP HANA node requires a redundant dedicated network connection with a minimum of 10Gb of bandwidth. Higher bandwidth is also supported. This network must extend end to end from the storage layer through network switching and computing up to the guest operating system hosted on VMware vSphere. In addition to the physical switching infrastructure, a VMware distributed switch (vDS) is used to provide adequate performance and manageability of network traffic at the hypervisor layer.



As shown in the preceding figure, each SAP HANA node uses a dedicated port group on the VMware distributed switch. This port group allows for enhanced quality of service (QoS) and dedicated assignment of physical network interface cards (NICs) on the ESX hosts. To use dedicated physical NICs while preserving HA capabilities in the event of NIC failure, the dedicated physical NIC is configured as an active uplink. Additional NICs are configured as standby uplinks in the teaming and failover settings of the SAP HANA port group. In addition, jumbo frames (MTU 9,000) must be enabled end to end on physical and virtual switches. In addition, turn off flow control on all ethernet ports used for storage traffic on servers, switches, and storage systems. The following figure shows an example of such a configuration.

LRO (large receive offload) must be turned off for interfaces used for NFS traffic. For all other network configuration guidelines, see the respective VMware best practices guides for SAP HANA.



Time synchronization

You must synchronize the time between the storage controllers and the SAP HANA database hosts. To do so, set the same time server for all storage controllers and all SAP HANA hosts.

Storage controller setup

This section describes the configuration of the NetApp storage system. You must complete the primary installation and setup according to the corresponding ONTAP setup and configuration guides.

Storage efficiency

Inline deduplication, cross-volume inline deduplication, inline compression, and inline compaction are supported with SAP HANA in an SSD configuration.

NetApp Volume Encryption

The use of NetApp Volume Encryption (NVE) is supported with SAP HANA.

Quality of Service

QoS can be used to limit the storage throughput for specific SAP HANA systems or other applications on a shared-use controller. One use case would be to limit the throughput of development and test systems so that they cannot influence production systems in a mixed setup.

During the sizing process, you should determine the performance requirements of a nonproduction system. Development and test systems can be sized with lower performance values, typically in the range of 20% to 50% of a production- system KPI as defined by SAP.

Starting with ONTAP 9, QoS is configured on the storage volume level and uses maximum values for throughput (MBps) and the amount of I/O (IOPS).

Large write I/O has the biggest performance effect on the storage system. Therefore, the QoS throughput limit

should be set to a percentage of the corresponding write SAP HANA storage performance KPI values in the data and log volumes.

NetApp FabricPool

NetApp FabricPool technology must not be used for active primary file systems in SAP HANA systems. This includes the file systems for the data and log area as well as the `/hana/shared` file system. Doing so results in unpredictable performance, especially during the startup of an SAP HANA system.

Using the “snapshot-only” tiering policy is possible as well as using FabricPool in general at a backup target such as a NetApp SnapVault or SnapMirror destination.

> Using FabricPool for tiering Snapshot copies at primary storage or using FabricPool at a backup target changes the required time for the restore and recovery of a database or other tasks such as creating system clones or repair systems. Take this into consideration for planning your overall lifecycle-management strategy and check to make sure that your SLAs are still being met while using this function.

FabricPool is a good option for moving log backups to another storage tier. Moving backups affects the time needed to recover an SAP HANA database. Therefore, the option “tiering-minimum-cooling-days” should be set to a value that places log backups, which are routinely needed for recovery, on the local fast storage tier.

Storage configuration

The following overview summarizes the required storage configuration steps. Each step is covered in detail in the subsequent sections. In this section, we assume that the storage hardware is set up and that the ONTAP software is already installed. Also, the connections between the storage ports (10GbE or faster) and the network must already be in place.

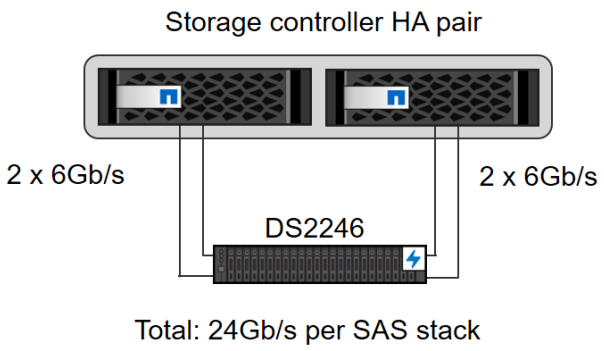
1. Check the correct disk shelf configuration as described in "Disk shelf connection."
2. Create and configure the required aggregates as described in "Aggregate configuration."
3. Create a storage virtual machine (SVM) as described in "SVM configuration."
4. Create LIFs as described in "LIF configuration."
5. Create volumes within the aggregates as described in "[Volume configuration for SAP HANA single host systems]" and "[Volume configuration for SAP HANA multiple host systems]."
6. Set the required volume options as described in "Volume options."
7. Set the required options for NFSv3 as described in "NFS configuration for NFSv3" or for NFSv4 as described in "NFS configuration for NFSv4."
8. Mount the volumes to namespace and set export policies as described in "Mount volumes to namespace and set export policies."

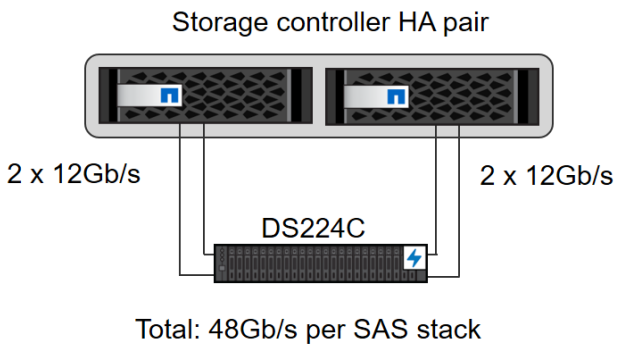
Disk shelf connection

SAS disk shelves

A maximum of one disk shelf can be connected to one SAS stack to provide the required performance for the SAP HANA hosts, as shown in the following figure. The disks within each shelf must be distributed equally to both controllers of the HA pair. ADPv2 is used with ONTAP 9 and the DS224C disk shelves.

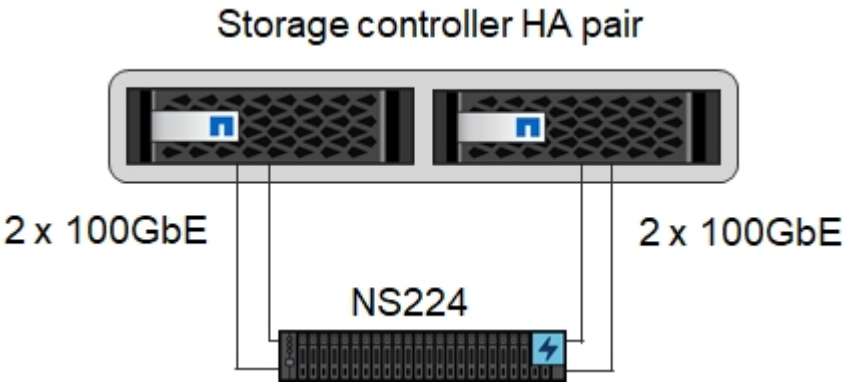
> With the DS224C disk shelf, quad-path SAS cabling can also be used but is not required.





NVMe (100GbE) disk shelves

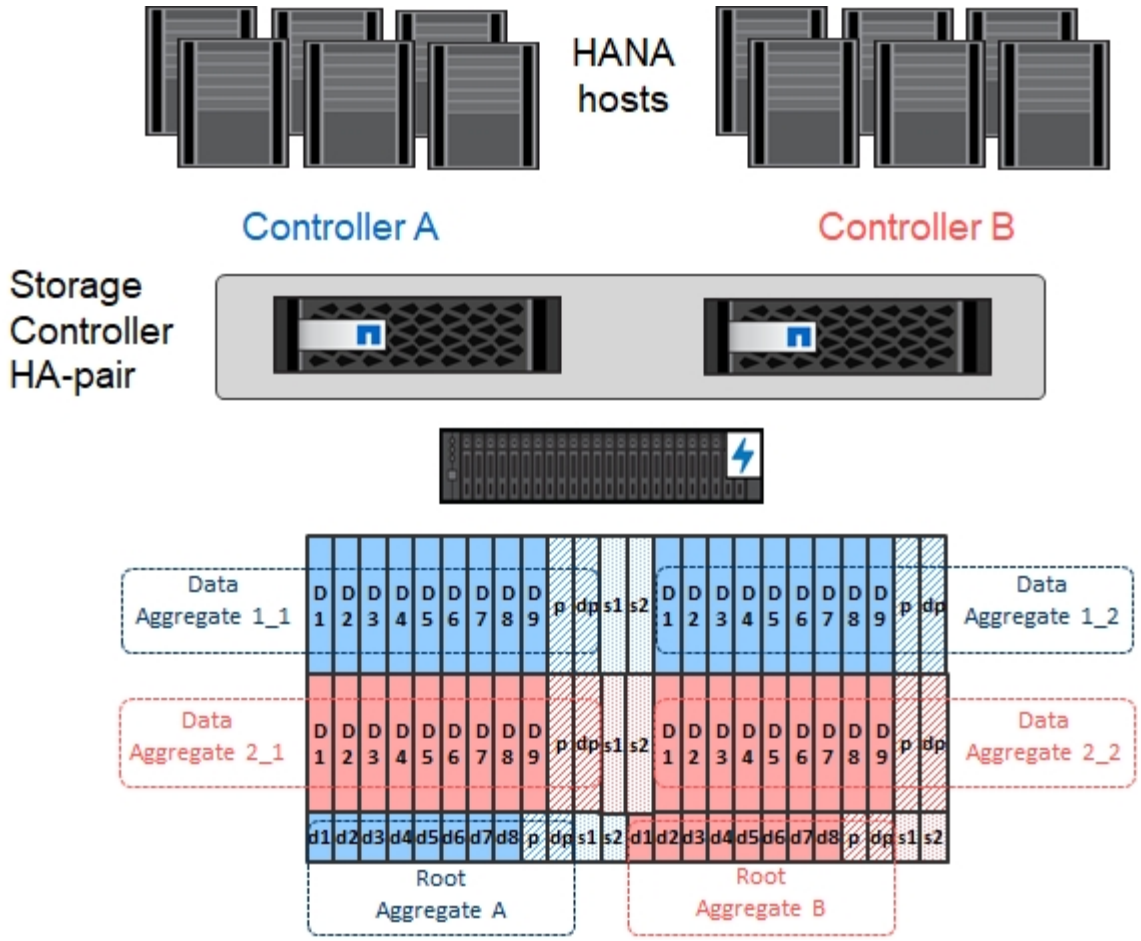
Each NS224 NVMe disk shelf is connected using two 100GbE ports per controller. The disks within each shelf must be distributed equally to both controllers of the HA pair. ADPv2, as described in the aggregate configuration chapter, is also used for the NS224 disk shelf. The following figure depicts the disk shelf connection with an NVMe drive.



Aggregate configuration

In general, you must configure two aggregates per controller, independent of the disk shelf or drive technology (SAS SSDs or NVMe SSDs) that is used. This step is necessary so that you can use all available controller resources. For AFF A200 series systems, one data aggregate is enough.

The following image shows a configuration of 12 SAP HANA hosts running on a 12Gb SAS shelf configured with ADPv2. Six SAP HANA hosts are attached to each storage controller. Four separate aggregates, two at each storage controller, are configured. Each aggregate is configured with 11 disks with nine data and two parity disk partitions. For each controller, two spare partitions are available.



SVM configuration

Multiple SAP landscapes with SAP HANA databases can use a single SVM. An SVM can also be assigned to each SAP landscape, if necessary, in case they are managed by different teams within a company.

If there is a QoS profile automatically created and assigned while creating a new SVM, remove this automatically created profile from the SVM to enable the required performance for SAP HANA:

```
vserver modify -vserver <svm-name> -qos-policy-group none
```

LIF configuration

For SAP HANA production systems, you must use different LIFs to mount the data volume and the log volume from the SAP HANA host. Therefore at least two LIFs are required.

The data and log volume mounts of different SAP HANA hosts can share a physical storage network port by either using the same LIFs or by using individual LIFs for each mount.

The maximum amount of data and log volume mounts per physical interface are shown in the following table.

Ethernet port speed	10GbE	25GbE	40GbE	100GeE
Maximum number of log or data volume mounts per physical port	2	6	12	24

> Sharing one LIF between different SAP HANA hosts might require a remount of data or log volumes to a different LIF. This change avoids performance penalties if a volume is moved to a different storage controller.

Development and test systems can use more data and volume mounts or LIFs on a physical network interface.

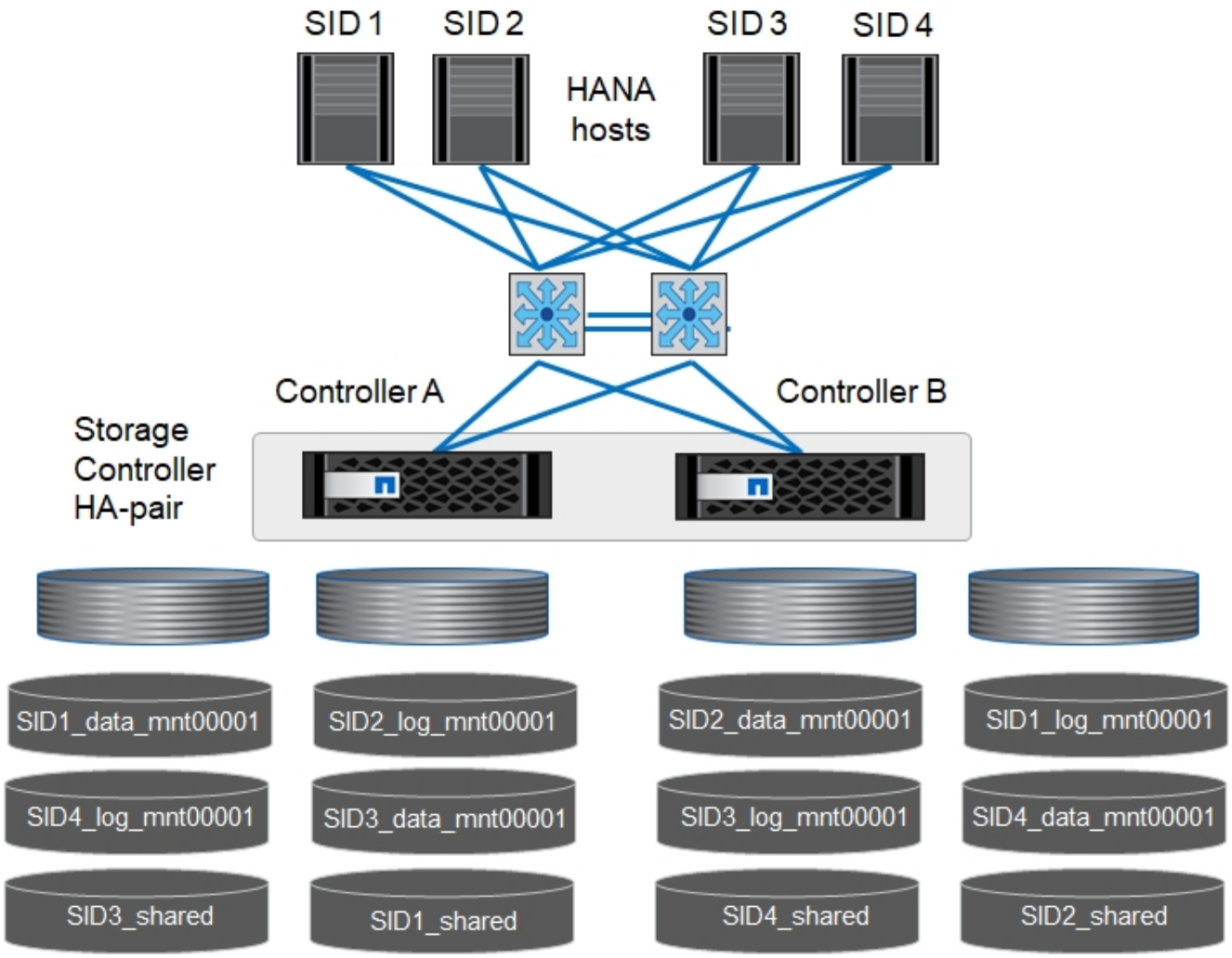
For production, development, and test systems, the `/hana/shared` file system can use the same LIF as the data or log volume.

Volume configuration for SAP HANA single-host systems

The following figure shows the volume configuration of four single-host SAP HANA systems. The data and log volumes of each SAP HANA system are distributed to different storage controllers. For example, volume `SID1_data_mnt00001` is configured on controller A, and volume `SID1_log_mnt00001` is configured on controller B.

> If only one storage controller of an HA pair is used for the SAP HANA systems, data and log volumes can also be stored on the same storage controller.

> If the data and log volumes are stored on the same controller, access from the server to the storage must be performed with two different LIFs: one LIF to access the data volume and the other to access the log volume.



For each SAP HANA host, a data volume, a log volume, and a volume for `/hana/shared` are configured. The following table shows an example configuration for single-host SAP HANA systems.

Purpose	Aggregate 1 at Controller A	Aggregate 2 at Controller A	Aggregate 1 at Controller B	Aggregate 2 at Controller b
Data, log, and shared volumes for system SID1	Data volume: SID1_data_mnt00001	Shared volume: SID1_shared	–	Log volume: SID1_log_mnt00001
Data, log, and shared volumes for system SID2	–	Log volume: SID2_log_mnt00001	Data volume: SID2_data_mnt00001	Shared volume: SID2_shared
Data, log, and shared volumes for system SID3	Shared volume: SID3_shared	Data volume: SID3_data_mnt00001	Log volume: SID3_log_mnt00001	–
Data, log, and shared volumes for system SID4	Log volume: SID4_log_mnt00001	–	Shared volume: SID4_shared	Data volume: SID4_data_mnt00001

The following table shows an example of the mount point configuration for a single-host system. To place the home directory of the `sidadm` user on the central storage, the `/usr/sap/SID` file system should be mounted

from the `SID_shared` volume.

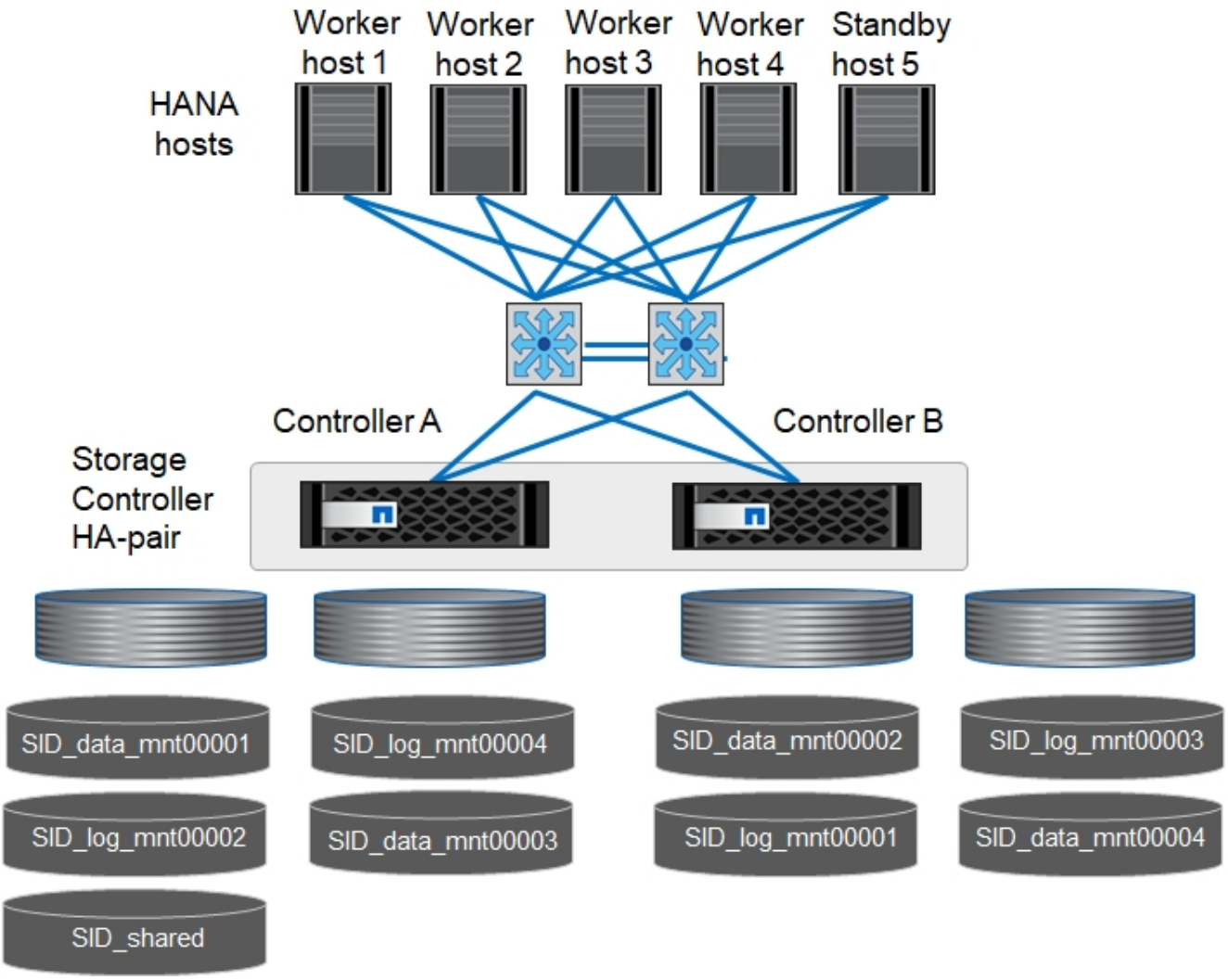
Junction path	Directory	Mount point at HANA host
SID_data_mnt00001		/hana/data/SID/mnt00001
SID_log_mnt00001		/hana/log/SID/mnt00001
SID_shared	usr-sap shared	/usr/sap/SID /hana/shared/

Volume configuration for SAP HANA multiple-host systems

The following figure shows the volume configuration of a 4+1 SAP HANA system. The data and log volumes of each SAP HANA host are distributed to different storage controllers. For example, volume `SID1_data1_mnt00001` is configured on controller A, and volume `SID1_log1_mnt00001` is configured on controller B.

If only one storage controller of an HA pair is used for the SAP HANA system, the data and log volumes can also be stored on the same storage controller.

If the data and log volumes are stored on the same controller, access from the server to the storage must be performed with two different LIFs: one LIF to access the data volume and one to access the log volume.



For each SAP HANA host, a data volume and a log volume are created. The `/hana/shared` volume is used by all hosts of the SAP HANA system. The following table shows an example configuration for a multiple-host SAP HANA system with four active hosts.

Purpose	Aggregate 1 at controller A	Aggregate 2 at controller A	Aggregate 1 at controller B	Aggregate 2 at controller B
Data and log volumes for node 1	Data volume: SID_data_mnt00001	–	Log volume: SID_log_mnt00001	–
Data and log volumes for node 2	Log volume: SID_log_mnt00002	–	Data volume: SID_data_mnt00002	–
Data and log volumes for node 3	–	Data volume: SID_data_mnt00003	–	Log volume: SID_log_mnt00003
Data and log volumes for node 4	–	Log volume: SID_log_mnt00004	–	Data volume: SID_data_mnt00004
Shared volume for all hosts	Shared volume: SID_shared			

The following table shows the configuration and the mount points of a multiple-host system with four active SAP HANA hosts. To place the home directories of the `sidadm` user of each host on the central storage, the

`/usr/sap/SID` file systems are mounted from the `SID_shared` volume.

Junction path	Directory	Mount point at SAP HANA host	Note
SID_data_mnt00001	–	/hana/data/SID/mnt00001	Mounted at all hosts
SID_log_mnt00001	–	/hana/log/SID/mnt00001	Mounted at all hosts
SID_data_mnt00002	–	/hana/data/SID/mnt00002	Mounted at all hosts
SID_log_mnt00002	–	/hana/log/SID/mnt00002	Mounted at all hosts
SID_data_mnt00003	–	/hana/data/SID/mnt00003	Mounted at all hosts
SID_log_mnt00003	–	/hana/log/SID/mnt00003	Mounted at all hosts
SID_data_mnt00004	–	/hana/data/SID/mnt00004	Mounted at all hosts
SID_log_mnt00004	–	/hana/log/SID/mnt00004	Mounted at all hosts
SID_shared	shared	/hana/shared/SID	Mounted at all hosts
SID_shared	usr-sap-host1	/usr/sap/SID	Mounted at host 1
SID_shared	usr-sap-host2	/usr/sap/SID	Mounted at host 2
SID_shared	usr-sap-host3	/usr/sap/SID	Mounted at host 3
SID_shared	usr-sap-host4	/usr/sap/SID	Mounted at host 4
SID_shared	usr-sap-host5	/usr/sap/SID	Mounted at host 5

Volume options

You must verify and set the volume options listed in the following table on all SVMs. For some of the commands, you must switch to the advanced privilege mode within ONTAP.

Action	Command
Disable visibility of Snapshot directory	vol modify -vserver <vserver-name> -volume <volname> -snapdir-access false
Disable automatic Snapshot copies	vol modify –vserver <vserver-name> -volume <volname> -snapshot-policy none
Disable access time update, except of the SID_shared volume	set advanced vol modify -vserver <vserver-name> -volume <volname> -atime-update false set admin

NFS configuration for NFSv3

The NFS options listed in the following table must be verified and set on all storage controllers. For some of the commands shown in this table, you must switch to the advanced privilege mode.

Action	Command
Enable NFSv3	nfs modify -vserver <vserver-name> v3.0 enabled

Action	Command
ONTAP 9: Set NFS TCP maximum transfer size to 1MB	set advanced nfs modify -vserver <vserver_name> -tcp-max-xfer -size 1048576 set admin
ONTAP 8: Set NFS read and write size to 64KB	set advanced nfs modify -vserver <vserver-name> -v3-tcp-max-read -size 65536 nfs modify -vserver <vserver-name> -v3-tcp-max-write -size 65536 set admin

NFS configuration for NFSv4

The NFS options listed in the following table must be verified and set on all SVMs.

For some of the commands in this table, you must switch to the advanced privilege mode.

Action	Command
Enable NFSv4	nfs modify -vserver <vserver-name> -v4.1 enabled
ONTAP 9: Set NFS TCP maximum transfer size to 1MB	set advanced nfs modify -vserver <vserver_name> -tcp-max-xfer-size 1048576 set admin
ONTAP 8: Set NFS read and write size to 64KB	set advanced nfs modify -vserver <vserver_name> -tcp-max-xfer-size 65536 set admin
Disable NFSv4 access control lists (ACLs)	nfs modify -vserver <vserver_name> -v4.1-acl disabled
Set NFSv4 domain ID	nfs modify -vserver <vserver_name> -v4-id-domain <domain-name>
Disable NFSv4 read delegation	nfs modify -vserver <vserver_name> -v4.1-read -delegation disabled
Disable NFSv4 write delegation	nfs modify -vserver <vserver_name> -v4.1-write -delegation disabled
Disable NFSv4 numeric ids	nfs modify -vserver <vserver_name> -v4-numeric-ids disabled

For NFS version 4. 0, replace `4.1` with `4.0` in the previous commands. While NFSv4.0 is supported, NFSv4.1 is preferred.

The NFSv4 domain ID must be set to the same value on all Linux servers (`/etc/idmapd.conf`) and SVMs, as described in the section "SAP HANA installation preparations for NFSv4."

If you are using NFSV4.1, then pNFS is enabled and used by default (recommended).

Set the NFSv4 lease time at the SVM (as shown in the following table) if SAP HANA multiple host system are used.

Action	Command
Set the NFSv4 lease time	set advanced nfs modify -vserver <vserver_name> -v4-lease -seconds 10 set admin

Starting with HANA 2.0 SPS4, HANA provides parameters to control failover behavior. Instead of setting the lease time at the SVM level, NetApp recommends using these HANA parameters.

The parameters are within `nameserver.ini` as shown in the following table. Keep the default retry interval of 10 seconds within these sections.

Section within nameserver.ini	Parameter	Value
failover	normal_retries	9
distributed_watchdog	deactivation_retries	11
distributed_watchdog	takeover_retries	9

Mount volumes to namespace and set export policies

When a volume is created, the volume must be mounted to the namespace. In this document, we assume that the junction path name is the same as the volume name. By default, the volume is exported with the default policy. The export policy can be adapted if required.

Host setup

All the host-setup steps described in this section are valid for both SAP HANA environments on physical servers and for SAP HANA running on VMware vSphere.

Configuration parameter for SUSE Linux Enterprise Server

Additional kernel and configuration parameters at each SAP HANA host must be adjusted for the workload generated by SAP HANA.

SUSE Linux Enterprise Server 12 and 15

Starting with SUSE Linux Enterprise Server 12 SP1, the kernel parameter must be set in a configuration file in the `/etc/sysctl.d` directory. For example, you must create a configuration file with the name `91-NetApp-HANA.conf`.

```
net.core.rmem_max = 16777216
net.core.wmem_max = 16777216
net.ipv4.tcp_rmem = 4096 131072 16777216
net.ipv4.tcp_wmem = 4096 16384  16777216
net.core.netdev_max_backlog = 300000
net.ipv4.tcp_slow_start_after_idle=0
net.ipv4.tcp_no_metrics_save = 1
net.ipv4.tcp_moderate_rcvbuf = 1
net.ipv4.tcp_window_scaling = 1
net.ipv4.tcp_timestamps = 1
net.ipv4.tcp_sack = 1
```

Saptune, included in SLES for SAP OS versions, can be used to set these values. For more information, see SAP Note 3024346 (requires SAP login).

If NFSv3 is used for connecting the storage, `sunrpc.tcp_max_slot_table_entries` must be set in `/etc/modprobe.d/sunrpc.conf`. If the file does not exist, you must first create it by adding the following line:

```
options sunrpc tcp_max_slot_table_entries=128
```

If the nconnect mount option is used, this value can be increased from 256 to 512.

Configuration parameters for Red Hat Enterprise Linux 7.2 or later

You must adjust additional kernel and configuration parameters at each SAP HANA host for the workload generated by SAP HANA.

If NFSv3 is used for connecting the storage, you must set the parameter `sunrpc.tcp_max_slot_table_entries parameter` in `/etc/modprobe.d/sunrpc.conf`. If the file does not exist, you must first create it by adding the following line:

```
options sunrpc tcp_max_slot_table_entries=128
```

If the nconnect mount option is used, this value can be increased from 256 to 512.

Starting with Red Hat Enterprise Linux 7.2, you must set the kernel parameters in a configuration file in the `/etc/sysctl.d` directory. For example, you must create a configuration file with the name `91-NetApp-HANA.conf`.

```
net.core.rmem_max = 16777216
net.core.wmem_max = 16777216
net.ipv4.tcp_rmem = 4096 131072 16777216
net.ipv4.tcp_wmem = 4096 16384  16777216
net.core.netdev_max_backlog = 300000
net.ipv4.tcp_slow_start_after_idle=0
net.ipv4.tcp_no_metrics_save = 1
net.ipv4.tcp_moderate_rcvbuf = 1
net.ipv4.tcp_window_scaling = 1
net.ipv4.tcp_timestamps = 1
net.ipv4.tcp_sack = 1
```

Create subdirectories in /hana/shared volume

The following examples show an SAP HANA database with SID=NF2.

To create the required subdirectories, take one of the following actions:

- For a single- host system, mount the `/hana/shared` volume and create the `shared` and `usr-sap` subdirectories.

```
sapcc-hana-tst-06:/mnt # mount <storage-hostname>:/NF2_shared /mnt/tmp
sapcc-hana-tst-06:/mnt # cd /mnt/tmp
sapcc-hana-tst-06:/mnt/tmp # mkdir shared
sapcc-hana-tst-06:/mnt/tmp # mkdir usr-sap
sapcc-hana-tst-06:/mnt/tmp # cd ..
sapcc-hana-tst-06:/mnt # umount /mnt/tmp
```

- For a multiple-host system, mount the `/hana/shared` volume and create the `shared` and the `usr-sap` subdirectories for each host.

  The example commands show a 2+1 multiple-host HANA system.

```
sapcc-hana-tst-06:/mnt # mount <storage-hostname>:/NF2_shared /mnt/tmp
sapcc-hana-tst-06:/mnt # cd /mnt/tmp
sapcc-hana-tst-06:/mnt/tmp # mkdir shared
sapcc-hana-tst-06:/mnt/tmp # mkdir usr-sap-host1
sapcc-hana-tst-06:/mnt/tmp # mkdir usr-sap-host2
sapcc-hana-tst-06:/mnt/tmp # mkdir usr-sap-host3
sapcc-hana-tst-06:/mnt/tmp # cd ..
sapcc-hana-tst-06:/mnt # umount /mnt/tmp
```

Create mount points

The following examples show an SAP HANA database with SID=NF2.

To create the required mount point directories, take one of the following actions:

- For a single-host system, create mount points and set the permissions on the database host.

```
sapcc-hana-tst-06:/ # mkdir -p /hana/data/NF2/mnt00001
sapcc-hana-tst-06:/ # mkdir -p /hana/log/NF2/mnt00001
sapcc-hana-tst-06:/ # mkdir -p /hana/shared
sapcc-hana-tst-06:/ # mkdir -p /usr/sap/NF2
sapcc-hana-tst-06:/ # chmod -R 777 /hana/log/NF2
sapcc-hana-tst-06:/ # chmod -R 777 /hana/data/NF2
sapcc-hana-tst-06:/ # chmod -R 777 /hana/shared
sapcc-hana-tst-06:/ # chmod -R 777 /usr/sap/NF2
```

- For a multiple-host system, create mount points and set the permissions on all worker and standby hosts. The following example commands are for a 2+1 multiple-host HANA system.
  - First worker host:

```
sapcc-hana-tst-06:~ # mkdir -p /hana/data/NF2/mnt00001
sapcc-hana-tst-06:~ # mkdir -p /hana/data/NF2/mnt00002
sapcc-hana-tst-06:~ # mkdir -p /hana/log/NF2/mnt00001
sapcc-hana-tst-06:~ # mkdir -p /hana/log/NF2/mnt00002
sapcc-hana-tst-06:~ # mkdir -p /hana/shared
sapcc-hana-tst-06:~ # mkdir -p /usr/sap/NF2
sapcc-hana-tst-06:~ # chmod -R 777 /hana/log/NF2
sapcc-hana-tst-06:~ # chmod -R 777 /hana/data/NF2
sapcc-hana-tst-06:~ # chmod -R 777 /hana/shared
sapcc-hana-tst-06:~ # chmod -R 777 /usr/sap/NF2
```

- Second worker host:

```
sapcc-hana-tst-07:~ # mkdir -p /hana/data/NF2/mnt00001
sapcc-hana-tst-07:~ # mkdir -p /hana/data/NF2/mnt00002
sapcc-hana-tst-07:~ # mkdir -p /hana/log/NF2/mnt00001
sapcc-hana-tst-07:~ # mkdir -p /hana/log/NF2/mnt00002
sapcc-hana-tst-07:~ # mkdir -p /hana/shared
sapcc-hana-tst-07:~ # mkdir -p /usr/sap/NF2
sapcc-hana-tst-07:~ # chmod -R 777 /hana/log/NF2
sapcc-hana-tst-07:~ # chmod -R 777 /hana/data/NF2
sapcc-hana-tst-07:~ # chmod -R 777 /hana/shared
sapcc-hana-tst-07:~ # chmod -R 777 /usr/sap/NF2
```

- Standby host:

```
sapcc-hana-tst-08:~ # mkdir -p /hana/data/NF2/mnt00001
sapcc-hana-tst-08:~ # mkdir -p /hana/data/NF2/mnt00002
sapcc-hana-tst-08:~ # mkdir -p /hana/log/NF2/mnt00001
sapcc-hana-tst-08:~ # mkdir -p /hana/log/NF2/mnt00002
sapcc-hana-tst-08:~ # mkdir -p /hana/shared
sapcc-hana-tst-08:~ # mkdir -p /usr/sap/NF2
sapcc-hana-tst-08:~ # chmod -R 777 /hana/log/NF2
sapcc-hana-tst-08:~ # chmod -R 777 /hana/data/NF2
sapcc-hana-tst-08:~ # chmod -R 777 /hana/shared
sapcc-hana-tst-08:~ # chmod -R 777 /usr/sap/NF2
```

Mount file systems

Different mount options must be used depending on the NFS version and ONTAP release. The following file systems must be mounted to the hosts:

- `/hana/data/SID/mnt0000*`
- `/hana/log/SID/mnt0000*`
- `/hana/shared`
- `/usr/sap/SID`

The following table shows the NFS versions that you must use for the different files systems for single-host and multiple-host SAP HANA databases.

File systems	SAP HANA single host	SAP HANA multiple hosts
/hana/data/SID/mnt0000*	NFSv3 or NFSv4	NFSv4
/hana/log/SID/mnt0000*	NFSv3 or NFSv4	NFSv4
/hana/shared	NFSv3 or NFSv4	NFSv3 or NFSv4
/usr/sap/SID	NFSv3 or NFSv4	NFSv3 or NFSv4

The following table shows the mount options for the various NFS versions and ONTAP releases. The common parameters are independent of the NFS and ONTAP versions.

> SAP LaMa requires the /usr/sap/SID directory to be local. Therefore, don’t mount an NFS volume for /usr/sap/SID if you are using SAP LaMa.

For NFSv3, you must switch off NFS locking to avoid NFS lock cleanup operations in case of a software or server failure.

With ONTAP 9, the NFS transfer size can be configured up to 1MB. Specifically, with 40GbE or faster connections to the storage system, you must set the transfer size to 1MB to achieve the expected throughput values.

Common parameter	NFSv3	NFSv4	NFSv4.1	NFS transfer size with ONTAP 9	NFS transfer size with ONTAP 8
rw, bg, hard, timeo=600, noatime	vers=3,nolock	vers=4,minorversion=0,lock	vers=4,minorversion=1,lock	rsize=1048576,wsize=1048576	rsize=65536,wsize=65536

To improve read performance with NFSv3, NetApp recommends that you use the `nconnect=n` mount option, which is available with SUSE Linux Enterprise Server 12 SP4 or later and RedHat Enterprise Linux (RHEL) 8.3 or later.

Performance tests showed that `nconnect=8` provides good read results. Log writes might benefit from a lower number of sessions, such as `nconnect=2`. Be aware that the first mount from an NFS server (IP address) defines the amount of sessions being used. Further mounts do not change this even if different values are used for nconnect.

For NFSv4, the nconnect option is supported by NetApp for NFSv4.1, starting with ONTAP 9.8. First NFS clients supporting nconnect with NFSv4.1 are available with SLES15SP2 and RHEL 8.3. For additional information check Linux vendor documentation.

The following example shows a single host SAP HANA database with SID=NF2 using NFSv3 and an NFS transfer size of 1MB. To mount the file systems during system boot with the `/etc/fstab` configuration file, complete the following steps:

1. Add the required file systems to the `/etc/fstab` configuration file.

```
sapcc-hana-tst-06:/ # cat /etc/fstab
<storage- vif-data01>:/NF2_data_mnt00001 /hana/data/NF2/mnt00001 nfs
rw,vers=3,hard,timeo=600,rsize=1048576,wsize=1048576, bg, noatime,nolock
0 0
<storage- vif-log01>:/NF2_log_mnt00001 /hana/log/NF2/mnt00001 nfs
rw,vers=3,hard,timeo=600,rsize=1048576,wsize=1048576, bg, noatime,nolock
0 0
<storage- vif-data01>:/NF2_shared/usr- sap /usr/sap/NF2 nfs
rw,vers=3,hard,timeo=600,rsize=1048576,wsize=1048576, bg, noatime,nolock
0 0
<storage- vif-data01>:/NF2_shared/shared /hana/shared nfs
rw,vers=3,hard,timeo=600,rsize=1048576,wsize=1048576, bg, noatime,nolock
0 0
```

2. Run `mount -a` to mount the file systems on all hosts.

The next example shows a multiple-host SAP HANA database with SID=NF2 using NFSv4.1 for data and log file systems and NFSv3 for the `/hana/shared` and `/usr/sap/NF2` file systems. An NFS transfer size of 1MB is used.

1. Add the required file systems to the `/etc/fstab` configuration file on all hosts.

The `/usr/sap/NF2` file system is different for each database host. The following example shows `/NF2_shared/usr- sap- host1`.

```
stlrx300s8-5:/ # cat /etc/fstab
<storage- vif-data01>:/NF2_data_mnt00001 /hana/data/NF2/mnt00001 nfs
rw, vers=4, minorversion=1,hard,timeo=600,rsize=1048576,wsize=1048576,
bg, noatime,lock 0 0
<storage- vif-data02>:/NF2_data_mnt00002 /hana/data/NF2/mnt00002 nfs rw,
vers=4, minorversion=1,hard,timeo=600,rsize=1048576,wsize=1048576, bg,
noatime,lock 0 0
<storage- vif-log01>:/NF2_log_mnt00001 /hana/log/NF2/mnt00001 nfs rw,
vers=4, minorversion=1,hard,timeo=600,rsize=1048576,wsize=1048576, bg,
noatime,lock 0 0
<storage- vif-log02>:/NF2_log_mnt00002 /hana/log/NF2/mnt00002 nfs rw,
vers=4, minorversion=1,hard,timeo=600,rsize=1048576,wsize=1048576, bg,
noatime,lock 0 0
<storage- vif-data02>:/NF2_shared/usr- sap- host1 /usr/sap/NF2 nfs
rw,vers=3,hard,timeo=600,rsize=1048576,wsize=1048576, bg, noatime,nolock
0 0
<storage- vif-data02>:/NF2_shared/shared /hana/shared nfs
rw,vers=3,hard,timeo=600,rsize=1048576,wsize=1048576, bg, noatime,nolock
0 0
```

2. Run `mount -a` to mount the file systems on all hosts.

SAP HANA installation preparations for NFSv4

NFS version 4 and higher requires user authentication. This authentication can be accomplished by using a central user management tool such as a Lightweight Directory Access Protocol (LDAP) server or with local user accounts. The following sections describe how to configure local user accounts.

The administration user `<sidadm>` and the `sapsys` group must be created manually on the SAP HANA hosts and the storage controllers before the installation of the SAP HANA software begins.

SAP HANA hosts

If it does not already exist, you must create the `sapsys` group on the SAP HANA host. Choose a unique group ID that does not conflict with the existing group IDs on the storage controllers.

The user `<sidadm>` is created on the SAP HANA host. A unique ID must be chosen that does not conflict with existing user IDs on the storage controllers.

For a multiple-host SAP HANA system, the user and group ID must be the same on all SAP HANA hosts. The group and user are created on the other SAP HANA hosts by copying the affected lines in `/etc/group` and `/etc/passwd` from the source system to all other SAP HANA hosts.

The NFSv4 domain must be set to the same value on all Linux servers and SVMs. Set the domain parameter "`Domain = <domain_name>`" in file `/etc/idmapd.conf` for the Linux hosts.

Enable and start the NFS idmapd service:

```
systemctl enable nfs-idmapd.service
systemctl start nfs-idmapd.service
```

The latest Linux kernels do not require this step. You can safely ignore warning messages.

Storage controllers

The user ID and group ID must be the same on the SAP HANA hosts and the storage controllers. The group and user are created by entering the following commands on the storage cluster:

```
vserver services unix-group create -vserver <vserver> -name <group name>
-id <group id>
vserver services unix-user create -vserver <vserver> -user <user name> -id
<user-id> -primary-gid <group id>
```

Additionally, set the group ID of the UNIX user root of the SVM to 0.

```
vserver services unix-user modify -vserver <vserver> -user root -primary
-gid 0
```

I/O stack configuration for SAP HANA

Starting with SAP HANA 1.0 SPS10, SAP introduced parameters to adjust the I/O behavior and optimize the database for the file and storage systems used.

NetApp conducted performance tests to define the ideal values. The following table lists the optimal values inferred from the performance tests.

Parameter	Value
max_parallel_io_requests	128
async_read_submit	on
async_write_submit_active	on
async_write_submit_blocks	all

For SAP HANA 1.0 versions up to SPS12, these parameters can be set during the installation of the SAP HANA database, as described in SAP note 2267798: Configuration of the SAP HANA Database During Installation Using hdbparam.

Alternatively, the parameters can be set after SAP HANA database installation by using the `hdbparam` framework.

```
nf2adm@sapcc-hana-tst-06:/usr/sap/NF2/HDB00> hdbparam --paramset
fileio.max_parallel_io_requests=128
nf2adm@sapcc-hana-tst-06:/usr/sap/NF2/HDB00> hdbparam --paramset
fileio.async_write_submit_active=on
nf2adm@sapcc-hana-tst-06:/usr/sap/NF2/HDB00> hdbparam --paramset
fileio.async_read_submit=on
nf2adm@sapcc-hana-tst-06:/usr/sap/NF2/HDB00> hdbparam --paramset
fileio.async_write_submit_blocks=all
```

Starting with SAP HANA 2.0, `hdbparam` was deprecated and the parameters were moved to `global.ini`. The parameters can be set using SQL commands or SAP HANA Studio. For more details, see SAP note 2399079: Elimination of hdbparam in HANA 2. The parameters can also be set within the global.ini as shown below:

```
nf2adm@stlrx300s8-6: /usr/sap/NF2/SYS/global/hdb/custom/config> cat
global.ini
…
[fileio]
async_read_submit = on
async_write_submit_active = on
max_parallel_io_requests = 128
async_write_submit_blocks = all
…
```

As of SAP HANA 2.0 SPS5, you can use the `setParameter.py` script to set the correct parameters:

```
nf2adm@sapcc-hana-tst-03:/usr/sap/NF2/HDB00/exe/python_support>
python setParameter.py
-set=SYSTEM/global.ini/fileio/max_parallel_io_requests=128
python setParameter.py -set=SYSTEM/global.ini/fileio/async_read_submit=on
python setParameter.py
-set=SYSTEM/global.ini/fileio/async_write_submit_active=on
python setParameter.py
-set=SYSTEM/global.ini/fileio/async_write_submit_blocks=all
```

SAP HANA data volume size

As the default, SAP HANA uses only one data volume per SAP HANA service. Due to the maximum file size limitation of the file system, NetApp recommends limiting the maximum data volume size.

To do so automatically, set the following parameter in `global.ini` in the section `[persistence]`:

```
datavolume_striping = true
datavolume_striping_size_gb = 8000
```

This creates a new data volume after the 8,000GB limit is reached. SAP note 240005 question 15 provides more information.

SAP HANA software installation

Install on a single-host system

SAP HANA software installation does not require any additional preparation for a single-host system.

Install on a multiple-host system

To install SAP HANA on a multiple-host system, complete the following steps:

1. Using the SAP `hdbclm` installation tool, start the installation by running the following command at one of the worker hosts. Use the `addhosts` option to add the second worker (`sapcc-hana-tst-07`) and the standby host (`sapcc-hana-tst-08`).

```
sapcc-hana-tst-06:/mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/HDB_LCM_LINUX_X86_64 # ./hdblcm --action=install
--addhosts=sapcc-hana-tst-07:role=worker,sapcc-hana-tst-08:role=standby



SAP HANA Lifecycle Management - SAP HANA Database 2.00.052.00.1599235305
************************************************************************



Scanning software locations...
Detected components:
    SAP HANA AFL (incl.PAL,BFL,OFL) (2.00.052.0000.1599259237) in
/mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/HDB_AFL_LINUX_X86_64/packages
    SAP HANA Database (2.00.052.00.1599235305) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/HDB_SERVER_LINUX_X86_64/server
    SAP HANA Database Client (2.5.109.1598303414) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/HDB_CLIENT_LINUX_X86_64/client
    SAP HANA Smart Data Access (2.00.5.000.0) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/SAP_HANA_SDA_20_LINUX_X86_64/packages
    SAP HANA Studio (2.3.54.000000) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/HDB_STUDIO_LINUX_X86_64/studio
    SAP HANA Local Secure Store (2.4.24.0) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/HANA_LSS_24_LINUX_X86_64/packages
```

```
    SAP HANA XS Advanced Runtime (1.0.130.519) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_RT_10_LINUX_X86_64/packages
    SAP HANA EML AFL (2.00.052.0000.1599259237) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/HDB_EML_AFL_10_LINUX_X86_64/packages
    SAP HANA EPM-MDS (2.00.052.0000.1599259237) in /mnt/sapcc-
share/software/SAP/HANA2SP5-52/DATA_UNITS/SAP_HANA_EPM-MDS_10/packages
    GUI for HALM for XSA (including product installer) Version 1
(1.014.1) in /mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACALMPIUI14_1.zip
    XSAC FILEPROCESSOR 1.0 (1.000.85) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACFILEPROC00_85.zip
    SAP HANA tools for accessing catalog content, data preview, SQL
console, etc. (2.012.20341) in /mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSAC_HRTT_20/XSACHRTT12_20341.zip
    XS Messaging Service 1 (1.004.10) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACMESSSRV04_10.zip
    Develop and run portal services for customer apps on XSA (1.005.1)
in /mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACPORTALSERV05_1.zip
    SAP Web IDE Web Client (4.005.1) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSAC_SAP_WEB_IDE_20/XSACSAPWEBIDE05_1.zip
    XS JOB SCHEDULER 1.0 (1.007.12) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACSERVICES07_12.zip
    SAPUI5 FESV6 XSA 1 - SAPUI5 1.71 (1.071.25) in /mnt/sapcc-
share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACUI5FESV671_25.zip
    SAPUI5 SERVICE BROKER XSA 1 - SAPUI5 Service Broker 1.0 (1.000.3) in
/mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACUI5SB00_3.zip
    XSA Cockpit 1 (1.001.17) in /mnt/sapcc-share/software/SAP/HANA2SP5-
52/DATA_UNITS/XSA_CONTENT_10/XSACXSACOCKPIT01_17.zip


SAP HANA Database version '2.00.052.00.1599235305' will be installed.

Select additional components for installation:

  Index | Components | Description

---------------------------------------------------------------------------
```

```
---------------------
  1     | all        | All components
  2     | server     | No additional components
  3     | client     | Install SAP HANA Database Client version
2.5.109.1598303414
  4     | lss        | Install SAP HANA Local Secure Store version
2.4.24.0
  5     | studio     | Install SAP HANA Studio version 2.3.54.000000
  6     | smartda    | Install SAP HANA Smart Data Access version
2.00.5.000.0
  7     | xs         | Install SAP HANA XS Advanced Runtime version
1.0.130.519
  8     | afl        | Install SAP HANA AFL (incl.PAL,BFL,OFL) version
2.00.052.0000.1599259237
  9     | eml        | Install SAP HANA EML AFL version
2.00.052.0000.1599259237
  10    | epmmds     | Install SAP HANA EPM-MDS version
2.00.052.0000.1599259237

Enter comma-separated list of the selected indices [3]: 2,3
Enter Installation Path [/hana/shared]:
```

2. Verify that the installation tool installed all selected components at all worker and standby hosts.

Adding additional data volume partitions

Starting with SAP HANA 2.0 SPS4, additional data volume partitions can be configured. This allows you to configure two or more volumes for the data volume of an SAP HANA tenant database and scale beyond the size and performance limits of a single volume.

> Using two or more individual volumes for the data volume is available for SAP HANA single-host and SAP HANA multiple-host systems. You can add additional data volume partitions at any time.

Enabling additional data volume partitions

To enable additional data volume partitions, add the following entry within `global.ini` by using SAP HANA Studio or Cockpit in the SYSTEMDB configuration.

```
[customizable_functionalities]
persistence_datavolume_partition_multipath = true
```

> Adding the parameter manually to the `global.ini` file requires the restart of the database.

Volume configuration for single-host SAP HANA systems

The layout of volumes for a single-host SAP HANA system with multiple partitions is like the layout for a system with one data volume partition but with an additional data volume stored on a different aggregate as the log volume and the other data volume. The following table shows an example configuration of an SAP HANA single-host system with two data volume partitions.

Aggregate 1 at controller A	Aggregate 2 at controller A	Aggregate 1 at controller B	Aggregate 2 at controller b
Data volume: SID_data_mnt00001	Shared volume: SID_shared	Data volume: SID_data2_mnt00001	Log volume: SID_log_mnt00001

The following table shows an example of the mount point configuration for a single-host system with two data volume partitions.

Junction path	Directory	Mount point at HANA host
SID_data_mnt00001	–	/hana/data/SID/mnt00001
SID_data2_mnt00001	–	/hana/data2/SID/mnt00001
SID_log_mnt00001	–	/hana/log/SID/mnt00001
SID_shared	usr-sap shared	/usr/sap/SID /hana/shared

You can create the new data volume and mount it to the namespace using either NetApp ONTAP System Manager or the ONTAP CLI.

Volume configuration for multiple-host SAP HANA systems

The layout of volumes is like the layout for a multiple- host SAP HANA system with one data volume partition but with an additional data volume stored on a different aggregate as log volume and the other data volume. The following table shows an example configuration of an SAP HANA multiple-host system with two data volume partitions.

Purpose	Aggregate 1 at controller A	Aggregate 2 at controller A	Aggregate 1 at controller B	Aggregate 2 at controller B
Data and log volumes for node 1	Data volume: SID_data_mnt00001	–	Log volume: SID_log_mnt00001	Data2 volume: SID_data2_mnt00001
Data and log volumes for node 2	Log volume: SID_log_mnt00002	Data2 volume: SID_data2_mnt00002	Data volume: SID_data_mnt00002	–
Data and log volumes for node 3	–	Data volume: SID_data_mnt00003	Data2 volume: SID_data2_mnt00003	Log volume: SID_log_mnt00003
Data and log volumes for node 4	Data2 volume: SID_data2_mnt00004	Log volume: SID_log_mnt00004	–	Data volume: SID_data_mnt00004

Purpose	Aggregate 1 at controller A	Aggregate 2 at controller A	Aggregate 1 at controller B	Aggregate 2 at controller B
Shared volume for all hosts	Shared volume: SID_shared	–	–	–

The following table shows an example of the mount point configuration for a single-host system with two data volume partitions.

Junction path	Directory	Mount point at SAP HANA host	Note
SID_data_mnt00001	–	/hana/data/SID/mnt00001	Mounted at all hosts
SID_data2_mnt00001	–	/hana/data2/SID/mnt00001	Mounted at all hosts
SID_log_mnt00001	–	/hana/log/SID/mnt00001	Mounted at all hosts
SID_data_mnt00002	–	/hana/data/SID/mnt00002	Mounted at all hosts
SID_data2_mnt00002	–	/hana/data2/SID/mnt00002	Mounted at all hosts
SID_log_mnt00002	–	/hana/log/SID/mnt00002	Mounted at all hosts
SID_data_mnt00003	–	/hana/data/SID/mnt00003	Mounted at all hosts
SID_data2_mnt00003		/hana/data2/SID/mnt00003	Mounted at all hosts
SID_log_mnt00003		/hana/log/SID/mnt00003	Mounted at all hosts
SID_data_mnt00004		/hana/data/SID/mnt00004	Mounted at all hosts
SID_data2_mnt00004	–	/hana/data2/SID/mnt00004	Mounted at all hosts
SID_log_mnt00004	–	/hana/log/SID/mnt00004	Mounted at all hosts
SID_shared	shared	/hana/shared/SID	Mounted at all hosts
SID_shared	usr-sap-host1	/usr/sap/SID	Mounted at host 1
SID_shared	usr-sap-host2	/usr/sap/SID	Mounted at host 2
SID_shared	usr-sap-host3	/usr/sap/SID	Mounted at host 3
SID_shared	usr-sap-host4	/usr/sap/SID	Mounted at host 4
SID_shared	usr-sap-host5	/usr/sap/SID	Mounted at host 5

You can create the new data volume and mount it to the namespace using either ONTAP System Manager or the ONTAP CLI.

Host configuration

In addition to the tasks described in the section "Host Setup," the additional mount points and `fstab` entries for the new additional data volume/s must be created and the new volumes must be mounted.

1. Create additional mount points.

- For a single-host system, create mount points and set the permissions on the database host:

```
sapcc-hana-tst-06:/ # mkdir -p /hana/data2/SID/mnt00001
sapcc-hana-tst-06:/ # chmod -R 777 /hana/data2/SID
```

- For a multiple-host system, create mount points and set the permissions on all worker and standby hosts.

  The following example commands are for a 2-plus-1 multiple-host HANA system.

  - First worker host:

```
sapcc-hana-tst-06:~ # mkdir -p /hana/data2/SID/mnt00001
sapcc-hana-tst-06:~ # mkdir -p /hana/data2/SID/mnt00002
sapcc-hana-tst-06:~ # chmod -R 777 /hana/data2/SID
```

  - Second worker host:

```
sapcc-hana-tst-07:~ # mkdir -p /hana/data2/SID/mnt00001
sapcc-hana-tst-07:~ # mkdir -p /hana/data2/SID/mnt00002
sapcc-hana-tst-07:~ # chmod -R 777 /hana/data2/SID
```

  - Standby host:

```
sapcc-hana-tst-07:~ # mkdir -p /hana/data2/SID/mnt00001
sapcc-hana-tst-07:~ # mkdir -p /hana/data2/SID/mnt00002
sapcc-hana-tst-07:~ # chmod -R 777 /hana/data2/SID
```

2. Add the additional file systems to the `/etc/fstab` configuration file on all hosts.

   See the following example for a single-host system using NFSv4.1:

```
<storage-vif-data02>:/SID_data2_mnt00001 /hana/data2/SID/mnt00001 nfs
rw,
vers=4minorversion=1,hard,timeo=600,rsize=1048576,wsize=1048576,bg,noati
me,lock 0 0
```

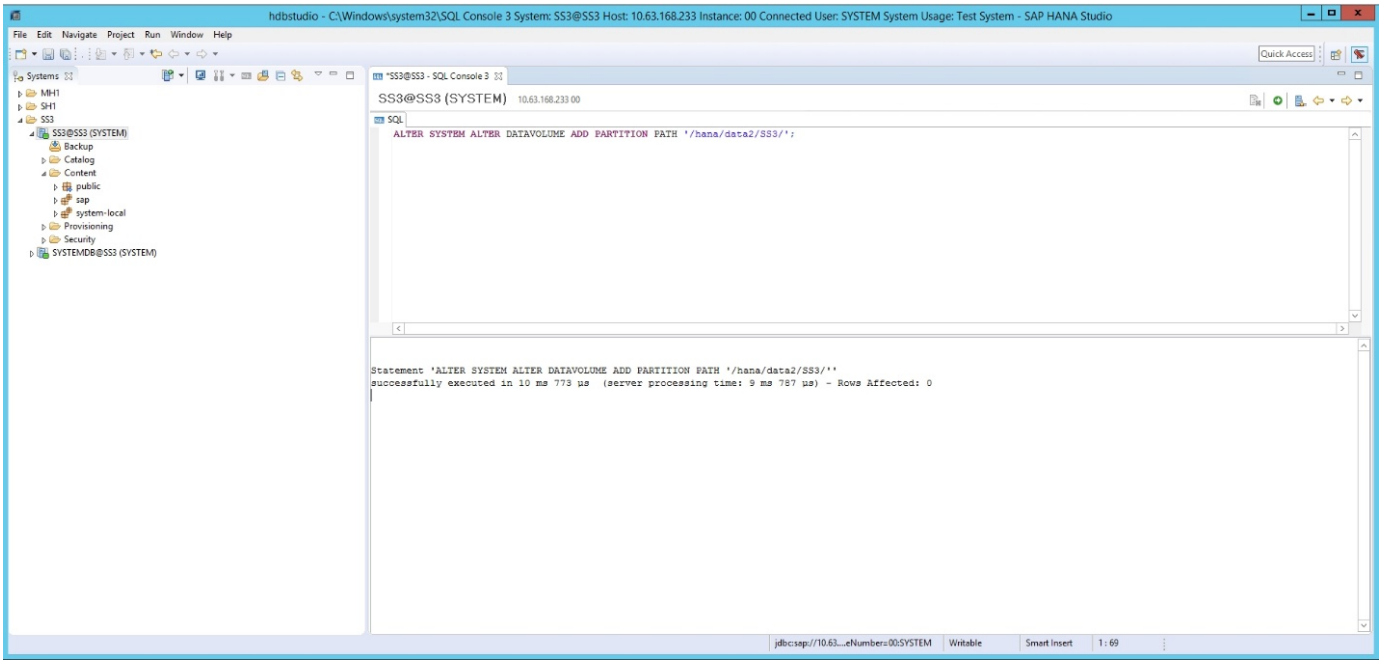
   Use a different storage virtual interface for connecting each data volume to ensure that you are using different TCP sessions for each volume

3. Mount the file systems by running the `mount -a` command.

Adding an additional data volume partition

Execute the following SQL statement against the tenant database to add an additional data volume partition to your tenant database. Use the path to additional volumes:

```
ALTER SYSTEM ALTER DATAVOLUME ADD PARTITION PATH '/hana/data2/SID/';
```



Where to find additional information

To learn more about the information described in this document, refer to the following documents and/or websites:

- Best Practices and Recommendations for Scale-Up Deployments of SAP HANA on VMware vSphere
  www.vmware.com/files/pdf/SAP_HANA_on_vmware_vSphere_best_practices_guide.pdf
- Best Practices and Recommendations for Scale-Out Deployments of SAP HANA on VMware vSphere
  http://www.vmware.com/files/pdf/sap-hana-scale-out-deployments-on-vsphere.pdf
- SAP Certified Enterprise Storage Hardware for SAP HANA
  http://www.sap.com/dmc/exp/2014-09-02-hana-hardware/enEN/enterprise-storage.html
- SAP HANA Storage Requirements
  http://go.sap.com/documents/2015/03/74cdb554-5a7c-0010-82c7-eda71af511fa.html
- SAP HANA Tailored Data Center Integration Frequently Asked Questions
  https://www.sap.com/documents/2016/05/e8705aae-717c-0010-82c7-eda71af511fa.html
- TR-4646: SAP HANA Disaster Recovery with Storage Replication
  https://www.netapp.com/us/media/tr-4646.pdf
- TR-4614: SAP HANA Backup and Recovery with SnapCenter
  https://www.netapp.com/us/media/tr-4614.pdf
- TR-4338: SAP HANA on VMware vSphere with NetApp FAS and AFF Systems
  www.netapp.com/us/media/tr-4338.pdf

- TR-4667: Automating SAP System Copies Using the SnapCenter 4.0 SAP HANA Plug- In
  https://www.netapp.com/us/media/tr-4667.pdf
- NetApp Documentation Centers
  https://www.netapp.com/us/documentation/index.aspx
- NetApp FAS Storage System Resources
  https://mysupport.netapp.com/info/web/ECMLP2676498.html
- SAP HANA Software Solutions
  www.netapp.com/us/solutions/applications/sap/index.aspx#sap-hana

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